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Gravity Device

Abstract

This disclosure relates to a dispensing system including: a display screen which displays one or more icons, the one or more icons including at least a first icon, a second icon, and a third icon, the display screen may transmit one or more signals to one or more processors based on one or more user contacts with the display screen, the one or more processors may receive at least a first sales data, a second sales data, and a third sales data, where the one or more processors may: modify a first size of the first icon based on a first sales data for a first element relating to the first icon; modify a second size of the second icon based on a second sales data for a second element relating to the second icon; and modify a third size of the third icon based on a third sales data for a third element relating to the third icon; and a first dispensing unit which may dispense one or more elements based on the one or more signals.

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Background/Summary

PRIORITY INFORMATION [0001] The present application claims priority to Provisional Patent Application No. 63/562,475 filed Mar. 7, 2024, which is incorporated in its entirety by reference.

BACKGROUND DISCUSSION
[0002] The dispensing industry is becoming more complex based on customer demand for customized drinks. These customized drinks require precision applications of various liquids and gases. In addition, material cost, labor cost, and labor safety are important factors that need to be enhanced. Further, customers demand a high level of cleaning to ensure that these customized drinks are safe. By utilizing this disclosure, the operator can achieve clean, customized, precision drinks with reduced material cost and labor cost while increase labor safety. In addition, customizable user interfaces can increase sales and customer satisfaction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] FIGS. **1**A-**1**D are illustrations of an exemplary embodiment of various user interfaces and marketing screens, according to one embodiment;
[0004] FIG. **2**A is an illustration depicting a maintenance screen/interface, according to one embodiment;
[0005] FIG. **2**B is an illustration of a customer utilizing a mobile device to interact with the dispensing device, according to one embodiment;
[0006] FIG. **2**C is an illustration of a drink selection process, according to one embodiment;
[0007] FIG. **2**D is an illustration of a drink management function, according to one embodiment;
[0008] FIG. **3** is an illustration of a sensor system utilizing in the dispensing device, according to one embodiment;
[0009] FIG. **4** is a block diagram of the dispensing system, according to one embodiment;
[0010] FIG. **5** is an illustration of a sold out condition, according to one embodiment;
[0011] FIG. **6** is an illustration of user interface with various drink data, according to one embodiment;
[0012] FIG. **7** is a block diagram of the dispensing system, according to one embodiment;
[0013] FIG. **8** is an illustration of the dispensing system, according to one embodiment;
[0014] FIG. **9** is a drink selection flowchart, according to one embodiment;
[0015] FIG. **10** is an illustration of the pressurized container containing a bag, according to one embodiment;
[0016] FIGS. **11**A-**11**D are illustrations of a customizable drink dispensing system, according to various embodiments;

[0017] FIG. **12** is an illustration of a bag in a pressurized dispensing system, according to one embodiment;
[0018] FIG. **13**A is a block diagram of a dispensing system, according to one embodiment;
[0019] FIG. **13**B is an illustration of a dispensing system, according to one embodiment;
[0020] FIGS. **14**A-**14**B are illustrations of drinking dispensing flowcharts, according to various embodiments;
[0021] FIGS. **15**A-**15**D are illustrations of drink dispensing systems, according to various embodiments;
[0022] FIGS. **16**A-**16**D are illustrations of drink dispensing systems, according to various embodiments;
[0023] FIG. **17** is an illustration of a CF Valve, according to one embodiment;
[0024] FIG. **18** is another illustration of a CF Valve, according to one embodiment;
[0025] FIG. **19** is an illustration of the drink dispensing device, according to one embodiment;
[0026] FIG. **20**A is a flowchart illustration of the customizable drink dispensing system, according to one embodiment;
[0027] FIG. **20**B is an illustration of a customizable drink dispensing device, according to one embodiment;
[0028] FIGS. **21**A-**21**D are illustrations of dispensing devices, according to various embodiments;
[0029] FIGS. **22**A-**22**D are illustrations of customizable dispensing devices and customizable cleaning systems, according to various embodiments;
[0030] FIGS. **23**A-**23**C are illustrations of attachment functions for dispensing devices, according to various embodiments;
[0031] FIGS. **24**A-**24**D are illustrations of the attachment device for the dispensing devices, according to various embodiments;
[0032] FIGS. **25**A-**25**B are illustrations of dispensing devices, according to various embodiments;
[0033] FIG. **25**C shows a block diagram of a dispensing device, according to one embodiment;
[0034] FIGS. **26**A-**26**B are illustrations of the attachment device, according to various embodiments;
[0035] FIGS. **27**A-**27**B are illustrations of a dispensing devices, according to various embodiments;
[0036] FIG. **28** is an illustration of an ice rack, according to one embodiment;
[0037] FIG. **29** is an illustration of a cooling system, according to one embodiment;
[0038] FIG. **30** is another illustration of a cooling system, according to one embodiment;
[0039] FIG. **31** is another illustration of a cooling system, according to one embodiment;
[0040] FIG. **32** is an illustration of an ice rack, according to one embodiment;
[0041] FIG. **33** is another illustration of an ice rack, according to one embodiment;
[0042] FIG. **34** is an illustration of a user interface for a dispensing system, according to one embodiment;
[0043] FIGS. **35**A-**35**B are illustrations of a CFiVe Switch cap, according to various embodiments;
[0044] FIG. **36** is an illustration of a dispensing system, according to one embodiment;
[0045] FIG. **37** is an illustration of a ventless CF Valve, according to one embodiment;
[0046] FIGS. **38**A-**38**B are illustrations of a ventless CF Valve, according to various embodiments;
[0047] FIG. **39** is an illustration of pressures which can be utilized with the ventless CF Valve; according to various embodiments;
[0048] FIGS. **40**A-**40**B are illustrations of a ventless CF Valve versus a vented CF Valve, according to various embodiments;
[0049] FIGS. **41**A-**41**B are illustrations of a ventless CF Valve versus a vented CF Valve, according to various embodiments;
[0050] FIGS. **42**A-**42**B are illustrations of a ventless CF Valve versus a vented CF Valve, according to various embodiments;
[0051] FIGS. **43**A-**43**B are illustrations of a ventless CF Valve versus a vented CF Valve, according

to various embodiments; and
[0052] FIGS. **44**A-**44**B are illustrations of a ventless CF Valve versus a vented CF Valve, according to various embodiments.
DETAILED DESCRIPTION
[0053] FIGS. **1**A-**1**D are illustrations of an exemplary embodiment of various user interfaces and marketing screens, according to one embodiment;
[0054] FIG. **1**A shows a display **102** with a first display screen **130** and a second display screen **132** (and/or Nth display screen). In one example, the first display screen **130** may include a first display item **108**, a second display item **110**, a third display item **112**, a fourth display item **114**, a fifth display item **116**, a sixth display item **118**, a seventh display item **120**, an eighth display item **122**, a nineth display item **124**, a tenth display item **126**, and an Nth display item **128**.
[0055] In various embodiments, the size of the icon (e.g., circle(s) in FIG. **1**A) corresponds with the amount of sales the specific drink has had over a specific time period. In one example, drink **1** (D**1**) has had the highest amount of sales and drink **11** (D**11**) has had the least amount of sales. This is represented by the icon for D**1** being the largest icon on the first display screen **130** and the icon for D**11** being the smallest icon on the first display screen **130**. In this example, the icons for D**2**, D**3**, D**4**, and D**5** have a medium size, which represents average sales volume. Whereas, the icons for D**6**, D**7**, D**8**, D**9**, D**10**, and D**11** have a small size, which represents below average sales volume. As previously stated, the icon for D**1** is the largest based on sales volume. In addition to the size of the icon changing, the location of the icon may be independently and/or in conjunction with a size change be changed based on sales data, time-of-day, a promotional, an event, and/or any other event disclosed in this document.
[0056] In another example, the second display screen **132** may include a first advertisement area **136** and a second advertisement area **138** (and/or an Nth advertisement area). Further, the second display screen **132** may include a first display item **134** and a first video area **140**.
[0057] In various embodiments, the first advertisement area **136**, the second advertisement area **138**, and/or first video area **140** may be utilized to advertise the bestselling product (D**1**) (and/or a worst selling product and/or an expiring product and/or a newly released product and/or any other criteria disclosed in this document) and/or any of the other drinking products and/or any other products (e.g., hot dog, chips, etc.). In one example, the first advertisement area **136** and the second advertisement area **138** are screen shots. Whereas, the first video area **140** is a streaming image. It should be noted that the first advertisement area **136** and the second advertisement area **138** may also be streaming videos and/or any other communication element. It should be noted that the display area may be on one screen and/or multiple screens.
[0058] In one example, a first dispensing area **104** is located at the bottom of the first display screen **130** and a second dispensing area **142** is located at the bottom of the second display screen **132**. In addition, a dispensing fluid **106** is shown. It should be noted that dispensing area may located anywhere on the dispensing machine and not directly coupled to the display area.
[0059] FIG. **1**B shows modified display item areas. In this example, the fifth display item **116** from FIG. **1**A usage has increasing making the fifth display item **116** the most popular (e.g., the highest number of purchases, the highest volume of purchases, etc.) and is now has the biggest display area. Therefore, the drink associated with the fifth display item **116** has moved to first display item area **144**. Further, the drink associated with the first display item area **108** from FIG. **1**A has moved to the second display item area **146** of FIG. **1**B because its usage and/or number of purchases and/or volume of purchases is less than the drink associated with the first display area **144**.
[0060] In an example, the most popular drink may be shown in a center display area **162** with a third advertisement **158** and/or an Nth advertisement **160** and/or an Nth video **164** displayed.
[0061] In various embodiments, one or more drinks may be ranked from most popular (e.g., highest sales) to least popular (e.g., lowest sales) and based on these rankings one or more icons may be assigned to one or more drinks (and/or products). For example, drink five (D**5**) has the fifth most

popular drink on FIG. **1**A. However, drink five (D**5**) has become the most popular drink and now has a large icon **144** associated with drink five. Whereas, drink one (D**1**) was the most popular drink on FIG. **1**A but now has become the second most popular drink and now has a second largest icon **146** associated with drink one. In one example, D**2**, D**3**, D**4**, D**6**, D**7**, D**8**, D**9**, D**10**, and D**11** have varying icon sizes (and/or locations) based on the amount of sales volume (and/or number of units sold). In this example, the third advertisement **158** and/or an Nth advertisement **160** and/or an Nth video **164** may be associated with drink five (D**5**) and D**5** displayed in the center of the display area as large icon **162**.

[0062] FIG. **1**C shows an event drink being displayed in the event drink area **172**, a fourth advertisement area **168**, a fifth advertisement area **170** (and/or Nth advertisement area), and a fourth video area **174** (and/or an Nth video area).

[0063] In various embodiments, a drink icon related to an event is shown as an event drink icon **172**. For example, a champion football game may have a first team and a second team playing in the champion football game. For a certain time period and/or geographical area and/or time of day and/or any other criteria, a drink event icon **172** may show a quarterback's favorite drink and/or drink combination (e.g., cherry soda). In another example, the circus may be in town and for a certain time period and/or geographical area and/or time of day and/or any other criteria, a drink event icon **172** may show an elephant drink, a lion drink, a clown drink, etc. In another example, a tournament (e.g., golf, poker, pool, bowling, etc.) may be in town and for a certain time period and/or geographical area and/or time of day and/or any other criteria, a drink event icon **172** may show an a golf player's favorite drink (e.g., the hole in one), a poker player's favorite drink (e.g., the straight flush), a pool player's favorite drink (e.g., the combo), and/or the bowling player's favorite drink (e.g., the strike and/or the turkey). In various examples, a fourth advertisement area **168**, a fifth advertisement area **170** (and/or Nth advertisement area), and a fourth video area **174** (and/or an Nth video area) may be utilized to support the event.

[0064] FIG. **1**D show a display **103** with a first display screen **131** and a second display screen **133**. In one example, display **103** may be one device, two device, and/or N devices. In one example, the first display screen **131** may include the first display item **108**, the second display item **110**, the third display item **112**, the fourth display item **114**, the fifth display item **116**, the sixth display item **118**, the seventh display item **120**, the eighth display item **122**, the nineth display item **124**, the tenth display item **126**, the Nth display item **128**, a first advertisement area **136**, a second (and/or Nth) advertisement area **138**, and/or a first video area **140**. In this example, the second display screen **133** may include a first display item **135**, the seventh display item **120**, a second display item **137**, the first display item **108** (from FIG. **1**A), the second display item **110** (from FIG. **1**A), a third display item **139**, a fourth display item **141**, a fifth display item **143**, the third display item **112** (from FIG. **1**A), a sixth display item **145**, a seventh display item **147**, an eighth display item **149**, a first advertisement **151**, a second (and/or Nth) advertisement **153**, and/or a Nth video stream **155**. In addition, one or more dispensing areas (**104** and **142**) are shown.

[0065] FIG. **2**A shows a maintenance display screen **200** with a maintenance overview area **202**. The maintenance overview area **202** may include a bag pressure **204**, a daily temperature and pressure routine **206**, a specific drink out **208**, a specific drink issue **210**, one or more drink issues **212**, and one or more temperature and/or pressure issues **214**.

[0066] In various embodiments, a maintenance screen **200** may include a first maintenance area **202**. First maintenance area **202** may have a bag pressure data area **204**, a daily temperature and pressure routine data area **206**, a specific drink out data area **208**, a specific drink data area **210**, one or more drink issues data area **212**, a temperature and/or pressure issues and/or any other maintenance data. In one embodiment, the bag pressure data area **204** may include one or more bag pressures which may be utilized to determine a sold out condition. In another example, one or more drinks may require a daily temperature procedure and/or pressure procedure which can be controlled via a controller utilizing data from the daily temperature and pressure routine data area

206. In another example, a first drink may be out for 4 hours while a second drink was out for 2 days which could be stored in the specific drink out data area **208**. Further, the cause of the drink out condition may be stored in the specific drink out data area **208**. In addition, one or more temperature and/or pressure issues may be stored and/or the solution to the issues may be stored in a temperature and/or pressure issues data area **214**.

[0067] FIG. **2**B shows one or more display screens **220** with one or more drink item areas, one or more advertisements, and/or one or more videos which can be communicated via the internet **222** to one or more remote devices **224** operated by one or more individuals **226**. Further, data relating to one or more drinks, maintenance, cleaning, promotions, sales, sports teams, etc. may be transmitted.

[0068] In various embodiments, the data relating to one or more drinks may be transmitted from the dispensing device **220** via a communication element (e.g., wired, wireless, and/or both) to one or more remote devices. Further, the user may preorder their drink selection which will be ready to be dispensing once the user is at the dispensing device.

[0069] FIG. **2**C shows a user interface **201** on a dispensing device. In one example, the user interface **201** may include one or more drink selection options **203**. In this example, the user selects a specific drink option **205**. Based on the specific drink option **205** being selected, a detailed drink selection option screen **207** is shown to the user. The detailed drink selection option screen **207** includes a brand area **209**, a main drink selection option **211** (e.g., Coke), a first added drink selection option **213** (e.g., cherry), a second added drink selection option **215** (e.g., vanilla), a third added drink selection option **217** (e.g., sugar free), a fourth added drink selection option **219** (e.g., lime), a fifth added drink selection option **221** (e.g., energy shot), and/or an Nth added drink selection option **223** (e.g., slurry—extra cold).

[0070] FIG. **2**D shows a first display screen **231** and a second display screen **233**. The first display screen **231** has a first plurality of drink options **237** and the second display screen **233** has a second plurality of drink options **235**. The first plurality of drink options **237** may be the same, completely different, and/or similar (e.g., some overlay) to the second plurality of drink options **235**. In one example, one or more of the drinking options in the first plurality of drink options **237** and/or the second plurality of drink options may be changed, modified, deleted, and/or added via a wireless **239** (and/or wired not shown) connection to a computer **241** and/or a mobile device **245** via a user **243** and/or an automated function.

[0071] FIG. **3** shows a dispensing system **300** with a first dispenser **302**, a second dispenser **304**, and/or an Nth dispenser **306**. In one example, the first dispenser **302** has a first sensor **310** which determines a first drink level **316**. Further, the second dispenser **304** has a second sensor **312** which determines a second drink level **318**. In addition, the Nth dispenser **306** has an Nth sensor **314** which determines a Nth drink level **320**. In this example, a drink flow **308** is determined via opening one or more valves of the first drink dispenser **302**, the second drink dispenser **304**, and/or the Nth drink dispenser **306**.

[0072] In various embodiments, one or more drinks may be created by opening valves on the first dispenser **302**, the second dispenser **304**, and/or the Nth dispenser **306** via a drink recipe data (e.g., open valve one for 1 second, valve 2 for 0.5 seconds, and valve N for 0.75 seconds. Further, one or more sensors may determine a time left before empty condition status. In other words, the first dispenser **302** has 1 hour of inventory left based on a first dispenser current usage rate, the second dispenser **304** has 2 days of inventory left based on a second dispenser current usage rate, and the Nth dispenser **306** has 7 days of inventory left based on an Nth dispenser current usage rate. Please note the even though the second dispenser **304** has a higher volume (e.g., **318** versus **320**) then the Nth dispenser **306**, the second dispenser **304** will run out of liquid sooner because its usage rate is higher than the Nth dispenser's usage rate.

[0073] FIG. **4** shows a dispensing device **400** which may include a bag-in-box **402** (and/or bag-in-bottle), a pump **404**, a pressure switch **406**, a valve **408**, a display **410**, a controller **412**, one or more remote device **414**, and/or one or more communication connections (e.g., **416**, **418**, **420**, etc.).

[0074] In various embodiments, the pressure switch **406** may obtain a sold out status because the pressure from the pump **404** is below a threshold amount. In this condition, the pressure switch **406** may send a signal to the display **410**, the controller **412**, and/or one or more remote devices **414** that a sold out condition exists. In addition, the drink will not be allowed to be dispensed.

[0075] FIG. **5** shows a first display status **502** with a first drink item area **506** and a second display status **504** with a first shaded drink item area **508**. In this embodiment, the first status **502** is that the first drink associated with the first drink item area **506** is available to be produced which is indicated by a lit up icon. Whereas, the second status **504** is that the first drink associated with the first drink item is unavailable based on a shaded icon **508**.

[0076] FIG. **6** shows a drink item area **602**. Drink item area **602** may include data including a carbonation, a sugar status, a recipe, a seasonal status, a mixing status, and/or a time of day information which are included in the data area **604**.

[0077] In various embodiments, a drink data area **600** may include one or more drink criteria data areas **602**. In this example, a first drink (D**1**) may be a carbonated drink, have sugar, have a first recipe, be included in a seasonal drink process, a mixture data, and/or a time-of-day information. Therefore, when a client selects a carbonated drink option and a sugar option, then D**1** would be displayed. However, if the client selects a carbonated drink option with a sugarless option, then D**1** would not be displayed. In addition, D**1** was selected, then A, B, and C drink options would not be available because drinks A, B, and C cannot be mixed with D**1**.

[0078] FIG. **7** shows a dispensing system **700** with a bag-in-box **702**, a pump **704**, a CFValve **706**, one or more communication elements (e.g., **708**, **710**, **712**, etc.), a controller **714**, a remote device **716**, a display **718**, and an Nth communication element **720**.

[0079] In various embodiments, the CFValve **706** may obtain a sold out status because the pressure from the pump **704** and/or any other pressure source is below a threshold amount. In this condition, the CFValve **706** may send a signal to the display **718**, the controller **714**, and/or one or more remote devices **716** that a sold out condition exists. In addition, the drink will not be allowed to be dispensed.

[0080] In FIG. **8**, an illustration of the dispensing system **800** is shown, according to one embodiment. In one example, the dispensing system **800** includes one or more small sized drink icons **802**, one or more medium sized drink icons **804**, one or more large sized drink icons **806**, one or more drink dispensing areas **808**, one or more ice dispensing areas **812**, and/or an ice tray **810**. In various examples, a drink option has an icon size and/or location based on the volume of sales (e.g., the larger the sales the larger the icon and the smaller the sales volume the smaller the icon), promotions (e.g., a larger icon based on the drink being promoted and/or on sale), specific theme (e.g., champion game-Gatorade may have a larger size), time of day (e.g., X drink is more popular in the morning whereas Y drink is more popular in the afternoon—Therefore, X drink has a larger icon in the morning and Y drink has a larger icon in the afternoon).

[0081] In FIG. **9**, a drink selection flowchart **900** is shown, according to one embodiment. In one example, the method **900** may include selecting a brand and/or special mix (step **902**). Further, the method **900** may include does the user want a flavor shot (and/or any other additive) (step **904**). If the answer is yes, then the method **900** may include selecting one or more flavor shots (and/or any other additive) (step **910**). The method **900** may include initiating a recipe based on the selections (step **906**) and the method may include dispensing the drink (step **908**). If the answer is no, then the method may initiate the recipe based on the selection (step **906**) and dispense the drink (step **908**).

[0082] In FIG. **10**, an illustration of a pressurized container containing a bag **1000** is shown, according to one embodiment. In one example, the pressurized container **1000** may include a container **1002**, a file cap **1004**, a pressure release device **1006**, a bag **1008**, a CF Valve **1010**, one or more mixing areas **1012**, and/or a dispensing area **1014**. In this example, the pressurized container **1000** may be utilized with any dispensing device and/or system disclosed in this

document.
[0083] In FIGS. **11**A-**11**D, illustrations of a customizable drink dispensing system are shown, according to various embodiments. FIG. **11**A shows a dispensing system **1100** including a housing **1102**, a sugar source **1104** (e.g., a first ingredient source), a non-sugar source **1106** (e.g., a second ingredient source), a first element source **1108** (e.g., a third ingredient source), a second element source **1110** (e.g., a fourth ingredient source), and/or an Nth element source **1112** (e.g., an Nth ingredient source). In one example, the dispensing system **1100** has a plurality of outlet areas (e.g., reference numbers **1114**, **1116**, **1118**, **1120**, **1122**, **1124**, **1126**, and/or **1128**). In one example, the plurality of outlets areas has two types of outlets: a) a multiple outlet area (e.g., a first multiple outlet area **1114** and an Nth multiple outlet area **1116**) and b) a single outlet area (e.g., a first single outlet area **1118**, a second single outlet area **1120**, a third single outlet area **1122**, a fourth single outlet area **1124**, a fifth single outlet area **1126**, and/or an Nth single outlet area **1128**). In one example, the multiple outlet areas (e.g., **1114** and **1116**) are outlet areas where two or more ingredient sources are dispensed to the multiple outlet area. In another example, the single outlet areas are outlet areas where only one ingredient is dispensed at a time. In various examples, utilizing various links (e.g., reference numbers **1130**, **1132**, **1134**, **1136**, **1138**, **1140**, **1142**, **1144**, **1146**, **1148**, **1150**, **1152**, and/or **1154**) a single ingredient may be dispensed to the single outlet areas (e.g., reference numbers **1118**, **1120**, **1122**, **1124**, **1126**, and/or **1128**) and/or two or more ingredients may be dispensed to the multiple outlet areas (e.g., reference numbers **1114** and **1116**).
[0084] In one example, a system may include multiple fluids that can be sauces, syrups, flavors, concentrated beverages, concentrated ingredients, non-concentrated beverages, ingredients and/or other fluids and/or gases. In this example, the system may dispense one or more liquids and/or gas from a single dispensing point. For example, certain fluids can be dispensed together from a multi-flavor dispense head and mixed as they are dispensed into a cup, pitcher, and/or downstream into a mixing chamber. In addition, a single fluid can be dispensed on their own from a single dispense point or through a multiple-dispense point.
[0085] In another example, the system may use a CF Valve to pressurize the dispensing which has the benefit of being faster than traditional pump bottles or systems that utilize pumps and plumbing. In addition, the CF Valve system is safer for employees because there are fewer repetitive motion injuries (e.g., from reaching up and pumping down especially with viscous liquids that require significant force on the pump to operate). In addition, there is substantially less waste because the product is delivered in flexible packaging compared to standard bottles/jugs (the volume of waste and trips to the dumpster can be reduced by up to 60%). Further, the CF Valve dispensing system is substantially more accurate than manual pumps which can be over +/−20 percent off in dispensing the amount due to differences in how an employee pumps the bottle. In addition, the system can be +/−5 percent accurate dispensing the product. In addition, the system can be cleaned easier than existing pump-based systems. Further, some fluids can be concentrated which reduces space requirements for operations and storage. This also reduces the amount of changeover required because the material lasts longer.
[0086] In FIG. **11**B, a dispensing system **1156** includes one or more pressure vessels that are utilized to control the flow of one or more elements. The dispensing system **1156** includes a housing **1158** (with one or more pressure vessels), a first syrup source **1160**, a second syrup source **1162**, an Nth syrup source **1164**, a multiple flavor source **1166** (with a plurality of flavors **1168**), and/or a dispensing area **1122** with dispensed products **1174** (from dispensing area **1172**). In one example, more than one bag may be in the pressure vessel. For example, a first ingredient and a second ingredient may be delivered from a first pressure vessel and a high ratio (30:1; 60:1; 100:1; etc.) element may be mixed from a second pressure vessel with either sugared or sugar-free syrup from a third pressure vessel all mixed together at a dispensing point and/or a mixing chamber.
[0087] FIG. **11**C shows a dispensing system **1176** including an order receiving device **1178**, a controller **1180**, a plurality of pressure vessel stacks **1182**, a first pressure vessel stack **1184**, a

second pressure vessel stack **1186**, a third pressure vessel stack **1188**, a fourth pressure vessel stack **1190**, an Nth pressure vessel stack **1192**, and/or a dispensing device **1194** with a dispensing area **1196**. In this example, the order receiving device **1178** may be a mobile computer and a stationary computing device. In one example, an order is entering into the order receiving device **1178** which is transmitted to the controller **1180**. In this example, the controller determines one or more recipes for the order and transmits one or more control signals to one or more of the plurality of pressure vessel stacks **1182** which generates the order and delivers the order via one or more pathways (e.g., reference numbers **1198**A, **1198**B, **1198**C, **1198**D, and/or **1198**E) to the dispensing device **1194** for dispensing into the dispensing area **1196**. In various examples, the one or more of the pressure vessel stacks in the plurality of pressure vessel stacks **1182** may have any number of pressure vessels (e.g., 1, 2, 3, 4, 5, 6, 11, 8, 9, 10, 11, . . . , 100). In addition, each pressure vessel may contain one element and/or ingredient. Further, each pressure vessel may contain more than one element and/or ingredient. For example, the first pressure vessel stack **1184** may contain a first stack with only a first syrup. In another example, the first pressure vessel stack **1184** may contain 10 stacks with each stack containing a single different syrup in each stack. In another example, the first pressure vessel stack **1184** may contain 10 stacks with 5 of the stacks containing a single different syrup in each stack and 5 of the stacks containing at least two different syrups in each stack. In another example, the second pressure stack **1186** may contain 20 stacks with 4 of the stacks containing a single different flavor in each stack; 6 stacks containing two different flavors in each stack; 3 stacks containing 5 different flavors in each stack, 2 stacks containing 3 different flavors in each stack; and 5 stacks containing 12 different flavors in each stack.

[0088] FIG. **11**D shows a dispensing system **1101** with a housing **1103**, one or more pressurized vessels (**1105** and **1105**A), and/or one or more fittings **11011**. In FIG. **8**, a dispensing system **800** with a pressure vessel **802**, a bag **804**, a fitting **806**, and outlet area **808**, and/or a pressure input area **810** is shown. In one example, the bag **804** is elongated and not completely filled with ingredients. In another example, the bag **804** has an outlet in the center of one end and a longer thinner shape to fit in the cylindrical canister (e.g., pressure vessel **802**). This allows for the optimization of the amount of fluid that can be put into the bag **804** and minimizes the labor required to exchange the bag **804** when emptied for a new bag. In one example, utilizing a rectangular bag with a cap/outlet on one corner can generate 30-40 percent wasted space inside of the pressure vessel **802**. By utilizing a longer thinner shape and having the outlet near the center, the wasted space is reduced to 10-15 percent.

[0089] In FIG. **12**, a dispensing device **1200** is shown. In one example, dispensing device **1200** includes a container **1202**, a bag **1204**, a bag outlet area **1206**, a dispensing area **1208**, and/or a container inlet area **1210**.

[0090] In FIGS. **13**A-**13**B, illustrations of block diagrams are shown, according to various embodiments. In FIG. **13**A, a block diagram is shown, according to one embodiment. A device **1300** may include a controller **1302**, one or more processors **1304**, one or more memories **1306**, one or more inventory modules **1308**, one or more maintenance modules **1310**, one or more cleaning modules **1312**, one or more drink dispensing modules **1314**, one or more loyalty card modules **1316**, one or more cameras **1318**, one or more sensors **1320**, one or more flavor modules **1322**, one or more number of actuations modules **1324**, one or more displays **1326**, one or more display modules **1328**, one or more time/day modules **1330**, and/or one or more transceivers **1332** and/or a recipe module **1358** (see FIG. **13**B).

[0091] In one example, the one or more inventory modules **1308** may be utilize to track one or more materials to be reordered. For example, material for creating a drink has reached a predetermined inventory level, therefore, the system automatically and/or via an approval function orders more of the material. In another example, one or more pressure vessel has been utilized a predetermined number of times, therefore, a maintenance request is issued and/or warning report generated via the one or more maintenance modules **1310**. In another example, a predetermined number of drinks have been created which requires a cleaning procedure to be initiated via the one or more cleaning modules **1312**. In another example, one or more drink dispensing modules **1314** may include one or more recipes for one or more drinks. In addition, the one or more loyalty card modules **1316** may be utilized to track a client's purchase and generate one or more rewards and/or discounts. In another example, the one or more cameras **1318** may be utilized to track system performance. In another example, the one or more sensors **1320** may be utilized to track the performance of one or more devices and/or fluid and/or elements. In addition, the one or more flavor modules **1322** may track data relating to one or more flavors. In addition, the one or more number of actuations modules **1324** may track the number of times a device is utilized. Further, the one or more displays **1326** may be utilized to display any data relating to this disclosure via the one or more display modules **1328**. In another example, the one or more time/day modules **1330** may be utilized to track purchase and/or activity rate throughout the day. In addition, the one or more transceivers **1332** may transmit any data in this disclosure to a remote device, control center, and/or any computing device. In addition, the recipe module **1358** may include any data relating to any recipe.

[0092] In another example, syrup control and/or management can be enhanced because dumping and/or walk away can be tracked. For example, when a person buys a fountain drink that person may take a slip and if the taste is not correct that person may dump the contents of the container and refill with another flavor. This might indicate that the syrup ratio is out of range and/or another quality control issue. In addition, the person may just walk away and not purchase anything which could be an indication that the syrup ratio is out of range and/or another quality control issue.

[0093] FIG. **13**B shows a drink delivery system **1350** including a drink input device **1352**, a display computer **1354**, a control board **1356**, a recipe module **1358**, a sauce board **1360**, a maintenance device **1362**, and/or a sauce controller(s) **1364**. In one example, an order is input via a touch screen (e.g., **1352**) which is then displayed on a computer (e.g., **1354**). Further, a control board **1356** communicates with the recipe module **1358** to determine which control signals to transmit to the sauce board **1360**. The sauce board **1360** transmits data to the sauce controller(s) **1364** which transmits signals to various elements to produce the drink. In addition, the sauce board **1360** transmits data to the maintenance device **1362** to track data and/or initiate any maintenance activities (e.g., cleaning, repairing, etc.).

[0094] In FIGS. **14**A-**14**C, illustrations of flowcharts are shown, according to various embodiments. In FIG. **14**A, a method **1400** may include receiving order(s) (step **1402**). The method **1400** may include determining criteria for the order(s) (step **1404**). The method **1400** may include initiating one or more control signals to one or more element devices (step **1406**). The method **1400** may include supplying one or more elements from one or more element devices to a dispensing area (step **1408**). The method **1400** may include providing data relating to one or more element devices, elements, orders, and/or sensor data to a control center and/or remote device (step **1410**). The method **1400** may include generating one or more reports, resupply orders, maintenance requests, sales data, time-of-day data, cost data, labor data, energy data, and/or promotional data to one or more of a control device, computing device, and/or remote computing device (step **1412**).

[0095] In FIG. **14**B, a method **1420** may include receiving order(s) (step **1422**). The method **1420** may include determining criteria for the order(s) (step **1424**). The method **1420** may include receiving one or more sensor data (e.g., pressure, humidity, temperature, flowrate, etc.) (step **1426**). The method **1420** may include modifying order criteria based on the received sensor data (step **1428**). The method **1420** may include initiating one or more control signals to one or more element devices (see FIG. **14**A—step **1406**). The method **1420** may include supplying one or more elements from one or more element devices to a dispensing area (step **1430**). The method **1420** may include providing data relating to one or more element devices, elements, orders, and/or sensor data to a control center and/or remote device (step **1432**). The method **1420** may include generating one or more reports, resupply orders, maintenance requests, sales data, time-of-day data, cost data, labor data, energy data, and/or promotional data to one or more of a control device, computing device, and/or remote computing device (step **1434**).

[0096] In FIGS. **15**A-**15**D, illustrations of the pressurized container dispensing system are shown, according to various embodiments. FIG. **15**A shows a pressurized dispensing system **1500** including the pressure vessel **1572**, the flexible bag **1574**, the first CF Valve **1576** with a solenoid (e.g., a CFiVe valve), the second CF Valve **1580**, the third solenoid **682**, the outlet area **684** (e.g., dispensing area), and/or the pressure input area **686**. In this example, the dispensing system **1500** has no electronic components on the outlet of the canister (e.g., pressure vessel **672**), which reduces the cost of the equipment, reduces cleaning requirements and cleaning time, and/or reduces the cost of installation, routine operations, and maintenance. In this example, the first CF Valve **676** with a solenoid pressurizes the pressure vessel **672** and is controlled by the inbound solenoid at an operating pressure of X (e.g., 27 PSI) and the second CF Valve **680** on the outlet of the canister operates at an operating pressure of Y (e.g., 15 PSI). In this example, the operating pressure of Y is less than the operating pressure of X. In one example, the first CF Valve **676** with a solenoid is opened and the canister is filled to the operating pressure X (e.g., 27 PSI) and as soon as the pressure in the pressure vessel **672** exceeds the operating pressure Y (e.g., 15 PSI) the second CF Valve **680** will open and dispense one or more elements (e.g., fluids, gas, etc.) from the flexible bag **674**. In various examples, the second CF Valve **680** may have a throttle pin and/or no throttle pin depending on the flow rate to be achieved by the one or more elements. In another example, the first CF Valve **676** with a solenoid will open for 0.83 seconds to dispense 0.42 ounces of a fluid at 27 PSI, once the pressure vessel **672** fluid (e.g., fluid in bag) exceeds 15 PSI the second CF Valve **680** will open and dispense the one or more elements.

[0097] FIG. **15**B shows a pressurized dispensing system **1502** including includes the pressure vessel **672**, the flexible bag **674**, the first CF Valve **676** with a solenoid (e.g., a CFiVe valve), the second solenoid **678**, the second CF Valve **680**, the outlet area **684** (e.g., dispensing area), the pressure input area **686**, and/or the pressure outlet area **688**. In this example, the dispensing system **1502** has no electronic components on the outlet of the canister (e.g., pressure vessel **672**), which reduces the cost of the equipment, reduces cleaning requirements and cleaning time, and/or reduces the cost of installation, routine operations, and maintenance. In this example, the first CF Valve **676** with a solenoid pressurizes the pressure vessel **672** and is controlled by the inbound solenoid at an operating pressure of X (e.g., 25 PSI) and the second CF Valve **680** on the outlet of the canister operates at an operating pressure of Y (e.g., 13 PSI). In this example, the operating pressure of Y is less than the operating pressure of X. In addition, the second solenoid **678** acts to depressurize the pressure vessel **672** when the dispensing operation is completed. In one example, the first CF Valve **676** with a solenoid is opened and the canister is filled to the operating pressure X (e.g., 25 PSI) and as soon as the pressure in the pressure vessel **672** exceeds the operating pressure Y (e.g., 13 PSI) the second CF Valve **680** will open and dispense one or more elements (e.g., fluids, gas, etc.) from the flexible bag **674**. In this example, when the appropriate amount of the one or more elements is dispensed, the second solenoid **678** will open to partially relieve the pressure from the pressure vessel **672** to under the threshold of the second CF Valve **680** operating pressure of Y. In various examples, the second CF Valve **680** may have a throttle pin and/or no throttle pin depending on the flow rate to be achieved by the one or more elements. In another example, the second solenoid **678** may be controlled with a set time to depressurize and/or with a pressure sensor to depressurize to a set pressure. There are many benefits to this dispensing system **1502**. For example, there are no electronics or electrical connections on the lid/cover of the pressure vessel **672** (which must be removed to replace the flexible bag **674** and to be cleaned). In another example, the first CF Valve **676** with a solenoid will open for 0.75 seconds to dispense 0.5 ounces of a fluid at 25 PSI, once the pressure vessel **672** fluid (e.g., fluid in bag) exceeds 13 PSI the second CF Valve **680** will open and dispense the one or more elements. Once the first CF Valve **676** with a solenoid closes, the second solenoid **678** will open for 0.6 seconds to relieve the pressure of fluid (e.g., fluid outside of the flexible bag **674**) in the pressure vessel **672** to 11 PSI and the second CF Valve **680** will close which stops the dispensing of the one or more elements (e.g., liquid, gas, etc.). In another example, the second solenoid **678** can be controlled with a pressure sensor so that the second solenoid **678** opens and then closes when the pressure in the pressure vessel **672** reaches a targeted closing pressure (e.g., 11 PSI in the above-referenced example).

[0098] In FIG. **15**C. a dispensing system **1506** includes the pressure vessel **672**, the flexible bag **674**, the first CF Valve **676** with a solenoid (e.g., a CFiVe valve), the second solenoid **678**, the third solenoid **682**, the outlet area **684** (e.g., dispensing area), the pressure input area **686**, and/or the pressure outlet area **688**. In this example, the dispensing system **1506** has no electronic components on the outlet of the canister (e.g., pressure vessel **672**), which reduces the cost of the equipment, reduces cleaning requirements and cleaning time, and/or reduces the cost of installation, routine operations, and maintenance. In this example, the first CF Valve **676** with a solenoid pressurizes the pressure vessel **672** and is controlled by the inbound solenoid at an operating pressure of X (e.g., 28 PSI) and the outlet of the canister operates at an operating pressure of Y (e.g., 16 PSI). In this example, the operating pressure of Y is less than the operating pressure of X. In addition, the second solenoid **678** acts to depressurize the pressure vessel **672** when the dispensing operation is completed. In one example, the first CF Valve **676** with a solenoid is opened and the canister is filled to the operating pressure X (e.g., 28 PSI) and as soon as the pressure in the pressure vessel **672** exceeds the operating pressure Y (e.g., 16 PSI) the third solenoid **682** will open and dispense one or more elements (e.g., fluids, gas, etc.) from the flexible bag **674**. In this example, when the appropriate amount of the one or more elements is dispensed, the second solenoid **678** will open to partially relieve the pressure from the pressure vessel **672** to under the threshold operating pressure of Y. In another example, the second solenoid **678** may be controlled with a set time to depressurize and/or with a pressure sensor to depressurize to a set pressure. There are many benefits to this dispensing system **1506**. For example, there are no electronics or electrical connections on the lid/cover of the pressure vessel **1572** (which must be removed to replace the flexible bag **1574** and to be cleaned). In another example, the first CF Valve **1576** with a solenoid will open for 0.63 seconds to dispense 0.75 ounces of a fluid at 28 PSI, once the pressure vessel **1572** fluid (e.g., fluid in bag) exceeds 16 PSI the third solenoid **1582** will open and dispense the one or more elements. Once the first CF Valve **1576** with a solenoid closes, the second solenoid **1578** will open for 0.9 seconds to relieve the pressure of fluid (e.g., fluid outside of the flexible bag **1574**) in the pressure vessel **1572** to 15 PSI and the second CF Valve **1580** will close which stops the dispensing of the one or more elements (e.g., liquid, gas, etc.). In another example, the second solenoid **1578** can be controlled with a pressure sensor so that the second solenoid **1578** opens and then closes when the pressure in the pressure vessel **1572** reaches a targeted closing pressure (e.g., 15 PSI in the above-referenced example).

[0099] In FIG. **15**D. a dispensing system **1510** includes a first disconnect device **1512**, a second disconnect device **1515**, a cartridge **1516** (e.g., filled with flavors, liquids, elements, ingredients, etc.), a third disconnect device **1518**, the first CF Valve **1576** with a solenoid (e.g., a CFiVe valve), the second solenoid **1578**, the second CF Valve **1580**, the third solenoid **1582**, the outlet area **1584** (e.g., dispensing area), the pressure input area **1586**, and/or the pressure outlet area **1588**. In this example, the dispensing system **1510** has no electronic components on the outlet of the canister (e.g., pressure vessel **1572**), which reduces the cost of the equipment, reduces cleaning requirements and cleaning time, and/or reduces the cost of installation, routine operations, and maintenance. In this example, the first CF Valve **1576** with a solenoid pressurizes the cartridge **1516** and is controlled by the inbound solenoid at an operating pressure of X (e.g., 27 PSI) and the second CF Valve **1580** on the outlet of the canister operates at an operating pressure of Y (e.g., 15 PSI). In this example, the operating pressure of Y is less than the operating pressure of X. In addition, the second solenoid **1578** acts to depressurize the cartridge **1516** when the dispensing operation is completed. In one example, the first CF Valve **1576** with a solenoid is opened and the cartridge **1516** is filled and/or pressure applied to the cartridge to the operating pressure X (e.g., 27 PSI) and as soon as the pressure exceeds the operating pressure Y (e.g., 15 PSI) the second CF Valve **1580** will open and dispense one or more elements (e.g., fluids, gas, etc.) from the cartridge **1516**. In this example, when the appropriate amount of the one or more elements is dispensed, the second solenoid **678** will open to partially relieve the pressure from the cartridge **1516** to under the threshold of the second CF Valve **1580** operating pressure of Y. In various examples, the second CF Valve **680** may have a throttle pin and/or no throttle pin depending on the flow rate to be achieved by the one or more elements. In another example, the second solenoid **1578** may be controlled with a set time to depressurize and/or with a pressure sensor to depressurize to a set pressure. There are many benefits to this dispensing system **1510**. For example, there are no electronics or electrical connections on the lid/cover of the pressure vessel **1572** (which must be removed to replace the flexible bag **1574** and to be cleaned). In another example, the first CF Valve **1576** with a solenoid will open for 0.83 seconds to dispense 0.42 ounces of a fluid at 27 PSI, once the cartridge **1516** fluid (e.g., fluid in cartridge) exceeds 15 PSI the second CF Valve **1580** will open and dispense the one or more elements. Once the first CF Valve **1576** with a solenoid closes, the second solenoid **1578** will open for 0.7 seconds to relieve the pressure on the cartridge **1516** to 12 PSI and the second CF Valve **1580** will close which stops the dispensing of the one or more elements (e.g., liquid, gas, etc.). In another example, the second solenoid **1578** can be controlled with a pressure sensor so that the second solenoid **1578** opens and then closes when the pressure in the cartridge **1516** reaches a targeted closing pressure (e.g., 12 PSI in the above-referenced example).

[0100] In FIG. **16**A, a dispensing system **1600** includes a pressure vessel **1602**, a flexible bag **1604**, a first pressure release valve **1606**, a first solenoid **1610**, a second pressure release valve **1608**, a second solenoid **1612**, an outlet area **1618** (e.g., dispensing area), a pressure input area **1614**, and/or a pressure outlet area **1616**. In FIG. **16**B, a dispensing system **1630** includes the pressure vessel **1602**, the flexible bag **1604**, the first pressure release valve **1606**, the second pressure release valve **1608**, the second solenoid **1612** (first solenoid in this example), the outlet area **1618** (e.g., dispensing area), and/or the pressure input area **1614**. In FIG. **16**C, a dispensing system **1650** includes the pressure vessel **1602**, the flexible bag **1604**, the first pressure release valve **1606**, the first solenoid **1610**, the second pressure release valve **1608**, the outlet area **1618** (e.g., dispensing area), the pressure input area **1614**, and/or the pressure outlet area **1616**.

[0101] In FIG. **16**D, the single CF Valve **1622** is providing a pressure function to the manifold **1624** of cartridges and/or pressure vessels. For example, the manifold **1624** is providing the first fluid flow **1626** to the first cartridge **1632** which generates the first fluid out **1638**. In addition, the manifold **1624** being fed by the CF Valve **1622** provides the second fluid flow **1628** to the second cartridge **1634** which generates the second fluid out **1640**. Further, the manifold **1624** provides the Nth fluid flow **1630** to the Nth cartridge **1636** which generates the Nth fluid out **1642**. In addition, the system may include a touch screen **1652** with one or more buttons **1654** or knobs **1662**. The touch screen **1652** may communicate with a control center **1656** to transmit one or more command signals to one or more cartridges and/or pressuring vessels, and/or CF Valves **1622** via one or more communication links (e.g., **1658**, **1660**, and **1661**). In addition, a sold-out device **1663** may be utilized.

[0102] FIGS. **17-18** are illustrations of a CF Valve, according to various embodiments. With reference to FIGS. **17-18**, a regulating valve in accordance with the present disclosure is generally depicted at **1710**. The valve includes an outer housing having a cap **1712** joined to a cup-shaped base **1714** at mating exterior flanges **1716**, **1718**.

[0103] The housing is internally subdivided by a barrier wall **1722** into a head section **1724** and a base section **1726**. An inlet **1728** in the cap **1712** is adapted to be connected to a fluid supply (not shown) having a pressure that can vary from below to above a threshold level. The inlet **1728** and a central port **1730** in the barrier wall **1722** are preferably aligned coaxially with a central axis A**1** of the valve. An outlet port **1731** is provided in the cap **1712**, and may be aligned on a second axis A**2** transverse to the first axis A**1**. Although the axis A**2** is shown at 90° with respect to axis A**1**, it will be understood that axis A**2** may be oriented at other angles with respect to axis A**1** in order to suit various applications of the valve.

[0104] A modulating assembly **1732** internally subdivides the base section into a fluid chamber **1723**′ segregated from a spring chamber **1723**″. The modulating assembly serves to prevent fluid flow through the valve when the fluid pressure at the inlet **1728** is below the threshold pressure. When the fluid pressure at the inlet exceeds the threshold pressure, the modulating assembly serves to accommodate fluid flow from the head section **1724** through port **1730** into fluid chamber **1723**′ and from there through outlet port **1731** at a substantially constant outlet pressure and flow rate. Either the outlet port **1731** or a downstream orifice or flow restrictor (not shown) serves to develop a back pressure in fluid chamber **1723**′.

[0105] The modulating assembly **1732** includes a piston comprised of a hollow shell **1734** and a central plug **1736**. The piston is supported for movement in opposite directions along axis A**1** by a flexible annular diaphragm **1738**. The inner periphery of the diaphragm is captured between the shell **1734** and plug **1736**. The cup shaped base **1714** has a cylindrical wall segment **1714**′ received within the cap **1712**. The outer periphery of the diaphragm is captured between an upper rim **1715** of the wall segment **1714**′ and an inwardly projecting interior ledge **1720** on the cap. The outer periphery of the diaphragm thus serves as an effective seal between the cap **1712** and base **1714**.

[0106] A stem **1740** on the piston plug **1736** projects through the port **1730** into the head section **1724**. An enlarged head **1742** on the stem has a tapered underside **1744** that coacts with a tapered surface **1746** of the barrier wall to modulate the size of the flow path through the port **1730** as an inverse function of the varying fluid pressure in the input section, with the result being to deliver fluid to the outlet **1731** at a substantially constant pressure and flow rate.

[0107] A compression spring **1748** in the spring chamber **1723**″ is captured between an underside surface of shell **1734** and the bottom wall **1752** of the housing base **1714**. The spring urges the modulating assembly **1732** towards the barrier wall **1722**. When the fluid inlet pressure is below the threshold pressure, spring **1748** serves to urge the diaphragm **1738** against a sealing ring **1749** on the underside of the barrier wall **1722**, thus preventing fluid through flow from the head section **1724** via port **1730** and fluid chamber **1723**′ to the outlet **1731**. As the fluid inlet pressure exceeds the threshold pressure, the resilient closure force of spring **1748** is overcome, allowing the modulating assembly to move away from the sealing ring **1749**, and allowing the modulating function of the coacting tapered surfaces **1744**, **1746** to commence. An opening **1750** in the bottom wall **1752** serves to vent the volume beneath diaphragm **1738** to the surrounding atmosphere.

[0108] In FIG. **19**, an illustration of an exemplary embodiment of a liquid delivery system is shown, according to one embodiment. FIG. **19** shows a dispensing device **1900** including a first flavor/component **1902**, a second flavor/component **1904**, an Nth flavor/component **1906**, a first CF Valve **1908** (e.g., a CF Valve only, a CF Valve with a Solenoid (e.g., CFIVe), and/or any other type of valve in this disclosure), a second CF Valve **19190**, an Nth CF Valve **1912**, a first solenoid **19194**, a second solenoid **1916**, an Nth solenoid **1918**, a first fixed orifice **1920**, a second fixed orifice **1922**, an Nth fixed orifice **1924**, a mixing vessel(s) **1926**, an output of the mixing vessel(s) **1928**, a post-mix area **1930**, an output of the post-mix area **1932**, an input device **1934**, and/or a controller **1936** with or without a recipe module **1938**.

[0109] In one example, the dispensing device is a recipe based system that is driven from a bank of two or more CFiVes (e.g., CF Valve and a Solenoid) that each represent a single fluid (liquid or gas) which then mix together to make a designated recipe. These can be pre-mix or post-mix (meaning they can mix in a manifold or vessel prior to dispense or mix at atmosphere at the point of dispense). In one example of a CF Valve application, the controlling orifice or flow insert after the outlet of the valve is changed in order to increase or decrease the total flow rate or amount poured. In contrast, the dispensing device **1900** shown in FIG. **19**, the flow rate is fixed with an orifice and the amount dispensed or mixed into the recipe is based on the "on time" designated in

the recipe.
[0110] For example, if a CFiVe with a specific orifice and a specific fluid flows at 1 ounce per second is utilized but the recipe only calls for 0.50 ounces, then the controller for the CFiVes will turn the CFiVe on and off again at a 50% duty cycle rate during a one second time slot to achieve the 0.50 ounces per second. Conversely, if the recipe calls for 2 ounces the controller will turn the CFiVe on and leave it actuated/open for 2 seconds to get the desired 2 ounces. The same ingredient can be dosed in different amounts for different recipes based on the “time on” dictated by the controller.
[0111] In legacy dispensers that use PRVs, ceramics or other types of flow control valves this level of control is not possible—meaning that if you want several different flow rates/amounts with the same ingredient you may require several separate valves for each flow rate imagined.
[0112] In this example, the system, the controller, and/or computer for the system has recipes (which are either entered into the equipment via flash drive, IOT download, manually, etc.) and there is a “library” of ingredients and flow rates per second for each ingredient through the CFiVe and the orifice. The system controller can turn on and off the various CFiVes for each ingredient for the allotted amount of time during the pour in order to achieve the targeted amount of each ingredient for that particular recipe. The system can be updated with additional ingredients and/or additional recipes.
[0113] The benefits of this system are that there is no need to visit the store/restaurant/equipment in order to change orifices to update flow rates. With a simple recipe update via internet download, flash drive or manual entry—the system can now run that recipe (flow rate/quantity) for each ingredient. Furthermore, if new ingredients are introduced, still there is no need for a service visit to the equipment as the information for that new ingredient is updated in the system and the system can use that ingredient in the updated recipes.
[0114] In FIGS. **20**A-**20**B, illustrations depicting various dispensing functions are shown, according to various embodiments. FIG. **20**A shows a dispensing device **2000** with two parts. The first part includes a CF Valve **2002**, a solenoid **2004**, and an aroma/essence element **2006**. The second part includes an area where dispensed ingredients are generated **2010** (e.g., coffee machine, soda machine, juice machine, etc.) which are then transported to an injection area **2008** where the aroma/essence element **2006** is injected into the dispensed ingredients and outputted to an output area **2012**.
[0115] In this system shown in FIG. **20**A, a CF Valve is used to provide a dispense system for a flavor shot or aroma/essence into a dispensed ingredient (beverage, sauce, syrup, condiment). One CF Valve is controlling the essence or highly concentrated flavor that will be dispensed at several dispense points into various drink concentrates or food ingredients such as sauces, syrups or condiments.
[0116] The benefit of this system shown in FIG. **20**A is that the essence or concentrated flavor will not lose its efficacy over time because it is kept in its purest form. For example, when mixed with syrups, sugar, sweetener and/or water, or other ingredients, the essence or concentrated flavor will become impure (e.g., contaminated) and the flavor and/or smell/essence will be reduced, which results in degraded beverage dispensing experiences.
[0117] In various examples, the system for the flavor dispenser can be a pressure dispensing system or a pump or atomizer to deliver the essence or concentrate flavor to the point of dispense.
[0118] FIG. **20**B shows a dispensing system **2050** including a CF Valve **2052**, a solenoid **2054**, a water dispenser **2056**, a carbonated water dispenser **2058**, an invert sugar dispenser **2060**, a corn syrup dispenser **2062**, and/or any other ingredient dispenser **2064**. In this example, the CF Valve **2052** maintains the pressure and/or flow rate for each and every dispenser (e.g., the water dispenser **2056**, the carbonated water dispenser **2058**, the invert sugar dispenser **2060**, the corn syrup dispenser **2062**, and/or the any other ingredient dispenser **2064**) to a water outlet **2066**, a carbonated water outlet **2068**, an invert sugar outlet **2070**, a corn syrup outlet **2072**, and/or any

other ingredient outlet **2074**.

[0119] Furthermore, in this system shown in FIG. **20**B, a single CF Valve can serve multiple dispensing heads for water, carb water, invert sugar, HF corn syrup, or any other ingredient. In this example, there is no need for multiple CF Valves at each point of dispense. So for example, in a carbonated drink machine there can be a single CF Valve may control the pressure and flow rates from one or more dispense points where the flavors, syrups, inclusions are mixed. In another example, one CF Valve can control invert sugar to multiple dispense point to be mixed with flavors, syrups and water. In another example, one CF Valve can control a condiment or sauce to be dispensed from one or more dispense points to be combined with other flavors, gasses, essences (example one source of Catsup can be mixed at one point of dispense with sriracha, and at another point be mixed with tabasco, and at another served plain).

[0120] In this system shown in FIG. **20**B, the benefit includes space savings and costs savings by utilizing one CF Valve to feed multiple points of dispense. This savings at the point of dispense allows for more flavors/dispense points to be fit into the same footprint making the dispensing equipment more effective for the user and store owner.

[0121] In FIGS. **21**A-**21**D, illustrations of an exemplary embodiment of a cleaning system are shown, according to one embodiment. FIG. **21**A shows a cleaning system **2100**, which includes a pressure source **2102**, a CF Valve **2104**, a container **2106**, a bag **2108**, a bag outlet line **2110**, a device **2112**, and a device outlet line **2116**. In this example, the CF Valve **2104** is a non-electric CF Valve.

[0122] In one example, the pressure in can be from city water, an air compressor, a pump, a compressed gas (e.g., CO.sub.2 and/or Nitrogen), and/or any other available pressure source. In this example, when the input fluid passes through the CF Valve, the CF Valve creates a constant pressure into the pressure canister **2106** of the input fluid (e.g., 7.5 PSI, 14 PSI, 21 PSI, 29 PSI, etc.) where that pressure will then act on the flexible package **2108** which contains the fluid (and/or cleaning element and/or ingredients (e.g., cleaner, sanitizer, etc.) and the fluid will be pushed through the flexible package **108** to the device **112** (and/or outlet).

[0123] The benefits of the cleaning system **2100** are that no pumps or plumbing or flow meters or connections are required to dispense any viscosity and/or any flow rate.

[0124] In another example, if the fluid viscosity is sensitive to temperature, a temperature sensor can be added to connect to the controllers of the equipment. If the temperature changes, the “on-time” of the CF Valves can be increased or decreased accordingly to account for the change in viscosity so that the same portion is dispensed to achieve a certain flow rate quantity to the point of cleaning.

[0125] FIG. **21**B shows a cleaning system **2120**, which includes the pressure source **2102**, the CF Valve **2104**, the container **2106**, the bag **2108**, the bag outlet line **2110**, the device **2112**, the device outlet line **2116** and a solenoid **2114**. In this example, the CF Valve **2104** is an electric CF Valve.

[0126] In one example, the pressure in can be from city water, an air compressor, a pump, a compressed gas (e.g., CO.sub.2 and/or Nitrogen), and/or any other available pressure source. In this example, when the input fluid passes through the CF Valve, the CF Valve creates a constant pressure into the pressure canister **2106** of the input fluid (e.g., 5.0 PSI, 10 PSI, 15 PSI, 20 PSI, etc.) where that pressure will then act on the flexible package **108** which contains the fluid (and/or cleaning element and/or ingredients (e.g., cleaner, sanitizer, etc.) and the fluid will be pushed through the flexible package **2108** to the device **2112** (and/or outlet).

[0127] The benefits of the cleaning system **2100** are that no pumps or plumbing or flow meters or connections are required to dispense any viscosity and/or any flow rate.

[0128] In another example, if the fluid viscosity is sensitive to temperature, a temperature sensor can be added to connect to the controllers of the equipment. If the temperature changes, the “on-time” of the CF Valves can be increased or decreased accordingly to account for the change in viscosity so that the same portion is dispensed to achieve a certain flow rate quantity to the point of

cleaning.
[0129] FIG. **21**C shows a cleaning system **2130**, which includes the pressure source **2102**, the CF Valve **2104**, the container **2106**, the bag **2108**, the bag outlet line **2110**, the device **2112**, the device outlet line **2116**, one or more sensors **2132**, a cleaning outlet line sensor **2136**, a device outlet line sensor **22138**, and a pressure relief device **2134**. In this example, the CF Valve **2104** is a non-electric CF Valve.
[0130] In one example, the one or more sensors **2132** may determine any characteristic of the pressure medium and based on the one or more determined medium characteristic modify any characteristic of the CF Valve **2104**, the bag **2108**, the bag outlet line **2110**, and/or any other system function and/or characteristic. In another example, the cleaning outlet line sensor **2136** may determine any characteristic of the medium in the bag outlet line **2110** and based on the one or more determined medium characteristic modify any characteristic of the CF Valve **2104**, the bag **2108**, the bag outlet line **2110**, and/or any other system function and/or characteristic. In another example, the device outlet line sensor **2138** may determine any characteristic of the medium in the device outlet line **2116** and based on the one or more determined medium characteristic modify any characteristic of the CF Valve **2104**, the bag **2108**, the bag outlet line **2110**, and/or any other system function and/or characteristic.
[0131] In one example, the pressure in can be from city water, an air compressor, a pump, a compressed gas (e.g., CO.sub.2 and/or Nitrogen), and/or any other available pressure source. In this example, when the input fluid passes through the CF Valve, the CF Valve creates a constant pressure into the pressure canister **2106** of the input fluid (e.g., 3.0 PSI, 9.0 PSI, 13.0 PSI, 18.0 PSI, etc.) where that pressure will then act on the bag **108** which contains the fluid (and/or cleaning element and/or ingredients (e.g., cleaner, sanitizer, etc.) and the fluid will be pushed through the flexible package **2108** to the device **2112** (and/or outlet).
[0132] The benefits of the cleaning system **2100** are that no pumps or plumbing or flow meters or connections are required to dispense any viscosity and/or any flow rate.
[0133] In another example, if the fluid viscosity is sensitive to temperature, a temperature sensor can be added to connect to the controllers of the equipment. If the temperature changes, the “on-time” of the CF Valves can be increased or decreased accordingly to account for the change in viscosity so that the same portion is dispensed to achieve a certain flow rate quantity to the point of cleaning.
[0134] FIG. **21**D shows a cleaning system **2140**, which includes the pressure source **102**, the CF Valve **2104**, the container **2106**, the bag **2108**, the bag outlet line **2110**, the device **2112**, the device outlet line **2116**, one or more sensors **2132**, the cleaning outlet line sensor **2136**, the device outlet line sensor **2138**, the pressure relief device **2134**, and the solenoid **2114**. In this example, the CF Valve **2104** is an electric CF Valve.
[0135] In one example, the one or more sensors **2132** may determine any characteristic of the pressure medium and based on the one or more determined medium characteristic modify any characteristic of the CF Valve **2104**, the bag **2108**, the bag outlet line **2110**, and/or any other system function and/or characteristic. In another example, the cleaning outlet line sensor **2136** may determine any characteristic of the medium in the bag outlet line **2110** and based on the one or more determined medium characteristic modify any characteristic of the CF Valve **2104**, the bag **2108**, the bag outlet line **2110**, and/or any other system function and/or characteristic. In another example, the device outlet line sensor **2138** may determine any characteristic of the medium in the device outlet line **2116** and based on the one or more determined medium characteristic modify any characteristic of the CF Valve **2104**, the bag **2108**, the bag outlet line **2110**, and/or any other system function and/or characteristic.
[0136] In one example, the pressure in can be from city water, an air compressor, a pump, a compressed gas (e.g., CO.sub.2 and/or Nitrogen), and/or any other available pressure source. In this example, when the input fluid passes through the CF Valve, the CF Valve creates a constant

pressure into the pressure canister **2106** of the input fluid (e.g., 10.0 PSI, 25.0 PSI, 45.0 PSI, 60.0 PSI, etc.) where that pressure will then act on the bag **2108** which contains the fluid (and/or cleaning element and/or ingredients (e.g., cleaner, sanitizer, etc.) and the fluid will be pushed through the flexible package **2108** to the device **2112** (and/or outlet).

[0137] In another example, if the fluid viscosity is sensitive to temperature, a temperature sensor can be added to connect to the controllers of the equipment. If the temperature changes, the "on-time" of the CF Valves can be increased or decreased accordingly to account for the change in viscosity so that the same portion is dispensed to achieve a certain flow rate quantity to the point of cleaning.

[0138] In FIG. **22**A, a cleaning system **2200** includes a water source **2202** (and/or a first pressure source), a second pressure source **2204**, a third pressure source **2206**, an Nth pressure source **2208**, a first CF Valve **2210**, a second CF Valve **2212**, a third CF Valve **2214**, an Nth CF Valve **2216**, a first container with a bag **2218**, a second container with a bag **2220**, an Nth container with a bag **2222**, a device or element **2224**, a controller **2226**, and/or one or more control lines **2228**. In one example, the first container with a bag **2218** has cleaning elements in the bag, the second container with a bag **2220** has sanitizer elements in the bag, and the Nth container with a bag **2220** has another element disclosed in this document.

[0139] In FIG. **22**B, a cleaning system **2250** includes the water source **2202** (and/or the first pressure source), the second pressure source **2204**, the third pressure source **2206**, the Nth pressure source **2208**, the first CF Valve **2210**, the second CF Valve **2212**, the third CF Valve **2214**, the Nth CF Valve **2216**, the first container with a bag **2218**, the second container with a bag **2220**, the Nth container with a bag **2222**, the device or element **2224**, the controller **2226**, the one or more control lines **2228**, a first check valve **2230**, a second check valve **2232**, a third check valve **2234**, and/or an Nth check valve **2236**. In one example, the first container with a bag **2218** has cleaning elements in the bag, the second container with a bag **2220** has sanitizer elements in the bag, and the Nth container with a bag **2220** has another element disclosed in this document.

[0140] In FIG. **22**C, a cleaning system **2260** includes the water source **2202** (and/or the first pressure source), the second pressure source **2204**, the third pressure source **2206**, the Nth pressure source **2208**, the first CF Valve **2210**, the second CF Valve **2212**, the third CF Valve **2214**, the Nth CF Valve **2216**, the first container with a bag **2218**, the second container with a bag **2220**, the Nth container with a bag **2222**, the controller **2226**, the one or more control lines **2228**, the first check valve **2230**, the second check valve **2232**, the third check valve **2234**, the Nth check valve **2236**, a pump **2262**, and/or one or more lines **2264** which lead to one or more devices and/or elements. In one example, the first container with a bag **2218** has cleaning elements in the bag, the second container with a bag **2220** has sanitizer elements in the bag, and the Nth container with a bag **2220** has another element disclosed in this document.

[0141] FIG. **22**D shows one or more cleaning recipes and/or procedures. In various embodiments, variables may be changed. For example, the time period, concentration of a cleaning element, sanitizer element, and/or any other element in this document, the order of operation, and/or any other characteristic may be modified and/or changed based on the targeting element to be cleaned and/or sanitized.

[0142] One of the benefits of this cleaning and/or sanitizing system is that any cleaner, or sanitizer, or cleaner/sanitizer can be utilizing in one or more canisters and then any number of chemicals and/or any amount of water (dilution) in any recipe to match any cleaning need. The same type of equipment could need different recipes depending on how it is used. For example, coffee machine that runs very oily beans gets cleaned with a higher concentration and more dwell times and more cycles of the cleaner. In another example, a coffee machine that runs not-oily can use a lower concentration and fewer cycles. In another example, a coffee machine can be cleaned daily or every **2250** or cups. So, a low volume store does not overuse chemicals. In another example, a drink machine that mixes concentrates with carbonated water or still water can also have a customized recipe based on what is being dispensed. For example, if a horchata drink tends to build up yeast or a dairy drink is more susceptible to bacteria then those can be on a more frequent and/or on a cycle that is longer has more cleaning steps and stronger concentrate. The same cannister of XYZ cleaner or ABC Sanitizer of DEF Cleaner/Sanitizer can be used in any number of applications by customizing the recipe and the fluidics downstream.

[0143] In one embodiment shown in FIG. **22**D, a first cleaning recipe is shown. The first cleaning recipe for drink dispenser number 1 may include: a first flush step (turn on CFValve #1 for 30 seconds); a first clean step (turn on CFValve #2 for 15 seconds); a first dwell step (turn on all CFValves off for 45 seconds); a first sanitize step (turn on CFValve #3 for 15 seconds); a second dwell step (turn all CFValves off for 45 seconds); and/or a second flush step (turn on CFValve #1 for 30 seconds).

[0144] In another embodiment shown in FIG. **22**D, a second cleaning recipe is shown. The second cleaning recipe for drink dispenser number 2 may include: a first flush step (turn on CFValve #1 for 15 seconds); a first clean step (turn on CFValve #2 for 20 seconds); a first dwell step (turn on all CFValves off for 60 seconds); a first sanitize step (turn on CFValve #3 for 30 seconds); and/or a second flush step (turn on CFValve #1 for 45 seconds).

[0145] In an nth embodiment shown in FIG. **22**D, an nth cleaning recipe is shown. The nth cleaning recipe for drink dispenser number N may include: a first flush step (turn on CFValve #1 for 20 seconds); a first clean step (turn on CFValve #2 for 25 seconds); a second flush step (turn on CFValve #1 for 40 seconds); a first sanitize step (turn on CFValve #3 for 45 seconds); a third flush step (turn on CFValve #1 for 60 seconds); a first hold step (turn off all CFValves for 30 seconds); and/or a fourth flush step (turn on CFValve #1 for 60 seconds).

[0146] In another embodiment shown in FIG. **22**D, a cleaning recipe is shown. The cleaning recipe for drink dispenser may include: a first flush step (turn on CFValve #1 for 40 seconds); a first clean/sanitize step (turn on CFValve #2 for 35 seconds); a first dwell step (turn all CFValves off for 60 seconds); and/or a second flush step (turn on CFValve #1 for 90 seconds).

[0147] In another embodiment shown in FIG. **22**D, a cleaning recipe is shown. The cleaning recipe for drink dispenser may include: a first flush step (turn on CFValve #1 for 50 seconds); a first/sanitize step (turn on CFValve #2 for 30 seconds); a first dwell step (turn all CFValves off for 120 seconds); and/or a second flush step (turn on CFValve #1 for 120 seconds).

[0148] In another embodiment shown in FIG. **22**D, a cleaning recipe is shown. The cleaning recipe for drink dispenser may include: a first flush step (turn on CFValve #1 for 40 seconds); a first clean/sanitize step (turn on CFValve #2 for 35 seconds); and/or a first dwell step (turn all CFValves off for 60 seconds.

[0149] In another embodiment shown in FIG. **22**D, a cleaning recipe is shown. The cleaning recipe for drink dispenser may include: a first clean step (turn on CFValve #2 for 50 seconds); a first dwell step (turn all CFValves off for 90 seconds); and/or a first flush step (turn on CFValve #1 for 30 seconds).

[0150] FIG. **23**A shows a system **2300** including a water source **2302**, an incoming water CF Valve **2304** (e.g., electric), a water line **2306**, an adapter **2308**, a fitting **2310**, a bag-in-bottle device **2312**, an outlet CF Valve **2314**, a faucet **2316**, a second incoming water CF Valve (e.g., electric) **2318**, an inlet cleaning line **2320**, a cleaning container **2322**, an outlet cleaning line **2324**, and/or an outlet cleaning line CF Valve **2326**.

[0151] FIG. **23**B shows a system **2330** including the water source **2302**, the incoming water CF Valve **2304** (e.g., electric), the water line **2306**, the adapter **2308**, the fitting **2310**, the bag-in-bottle device **2312**, the outlet CF Valve **2314**, the faucet **2316**, a second incoming water CF Valve (e.g., non-electric) **2332**, the inlet cleaning line **2320**, the cleaning container **2332**, the outlet cleaning line **2324**, and/or the outlet cleaning line CF Valve **2326**.

[0152] FIG. **23**C shows a system **2340** including a water/pressure source **2341**, a first CF Valve **2342**, a second CF Valve **2344**, a container **2346**, a bag **2348** (with cleaning solution, sanitizer solution, or both), the adapter **2308**, the fitting **2310**, a bag-in-bottle device **2350**, a third CF Valve **2352** (e.g., first flavor), a fourth CF Valve **2354** (e.g., second flavor), a fifth CF Valve **2356** (e.g., third flavor), an Nth CF Valve **2358** (e.g., Nth flavor), and/or a faucet **2360**.

[0153] In FIGS. **24**A-**24**D, illustrations of a fitting being utilized to connect the bag to the faucet are shown, according to various embodiments. In FIG. **24**A, a fitting **2430** includes an external end **2432**, a thread **2434**, an inside end **2436**, and/or an O-ring **2438**. In one example, the fitting **2430** is a connection device between the ingredients bag **2414** and the dispensing area **2418**. In this example, the fitting **2430** connects with the ingredients bag **2414** via the inside end **2436** entering the ingredients bag **2414**. In addition, the fitting **2430** connects with the dispensing area **2418** via the external end **2432**.

[0154] In FIG. **24**B, a fitting **2450** includes a lid **2452**, a faucet **2454**, a cleaning fixture/sleeve **2456**, a dry break **2458**, and/or a retaining clip **2460**. In this example, the CFive on the faucet side and/or dispensing side may connect to the bag inside the pressurized vessel with a connection fitting **2450**. In this example, the fitting **2450** connects the bag to dispensing area through the lid. In another example, the canister and/or lid can have an additional feature that holds the cap in the bag in place to make insertion of the connection directly into the bag consistent (e.g., no side loading).

[0155] In FIG. **24**C, a fitting **2462** includes a clip **2464** and a faucet outlet **2466**. In this example, the cleaning fixture can be connected to the inside of the lid for a clean-in-place functionality. In addition, the fitting **2462** will allow for cleaning and/or sanitizing fluid to clean the outside and inside of the fitting and to pass through the fitting and out the CF Valve by the faucet to clean all wetted surfaces. In FIG. **24**D, a fitting **2470** includes various internal elements **2472**, an O-ring **2474**, and/or a retaining clip **2476**. In one example, cleaning/sanitizing fluid cleans outside of probe fitting pas Cap/Bag O-ring seal (e.g., wetted surfaces). In another example, the O-ring in the sleeve seals to bottom of probe. In another example, the retaining clip keeps sleeve/fixture in place when pressurized.

[0156] FIG. **25**A shows a cleaning system **2500** which includes a first H.sub.2O source **2502**, a first CF Valve **2504**, a cleaning solution input area **2506**, a cleaning device **2508**, an inlet **2510** to a mixing area **2514**, which allows water to enter the mixing area **2514**, a cleaning concentrate bag **2516** where a non-diluted, pre-diluted, and/or full diluted cleaning concentrate enters the mixing area **2514** to mix with the water from inlet **2510**, the non-diluted, pre-diluted, and/or full diluted cleaning concentrate leaves the cleaning device **2508** via an exit line **2518** to a second CF Valve, which obtains a final mixture ratio by being mixed with the second water source **2522** and a third CF Valve **2524** to obtain cleaning solution output **2526**.

[0157] In various examples shown in FIG. **22**D, chemical compatibility, system material tolerances, Ph levels, drink materials, ratio requirements, and/or any other condition disclosed in this document, may be utilized to determine whether to non-dilute, pre-dilute, and/or fully dilute a cleaning system. For example, a 100 to 1 ratio; a 400 to 1 ratio; a 800 to 1 ratio; and/or a 1000 to 1 ratio may be utilized. In one example for coffee, a 100 to 1 ratio for the cleaning solution may be utilized.

[0158] FIG. **25**B shows a cleaning system **2540**, which includes a pressure source **2542**, a first CF Valve **2544**, a container **2546**, a bag **2548**, a bag outlet line **2550**, a second CF Valve and/or a first solenoid **2552**, a device **2554**, and a device outlet line **2556**. In this example, the CF Valves **2544** and **2552** may be non-electric CF Valves and/or electric CF Valves.

[0159] FIG. **25**C shows a block diagram **2560** which includes a camera(s) **2562**, sensor(s) **2564**, memory **2566**, processor(s) **2568**, a maintenance module(s) **2570**, cleaning module(s) **2572**, a transceiver **2574**, a dispensing database(s) **2576**, controller(s) **2578**, and/or device module(s) **2580**.

[0160] In one example, a maintenance module **2570** for a first dispensing system may include every drinking dispensing data (e.g., number of drinks, ratio, last maintenance cycle, temperature data, time of use data, etc.) for one or more dispensing devices.

[0161] In another example, a cleaning module **2572** may include different cleaning recipes for a dispensing device based on the dispensing device usage since the last cleaning cycle. For example, a large amount of high viscosity drinks since the last cleaning cycle may indicate that a higher concentrate cleaning solution, recipe, and/or procedure should be utilized. This information may be included for a plurality of dispensing devices in the dispensing database **2576**. Further, the controller **2578** may control one or more devices (CFValves, solenoids, check valves, etc.) to implement one or more cleaning recipes.

[0162] FIG. **26**A shows the fitting device **2611** and the adapter **2602** connected for the cleaning and/or sanitizing operation. In this example, the adapter **2602** has an inlet area **2604**, an outlet area **2606**, one or more walls **2608**, and a cleaning element area **2610**. In this example, water enters via the inlet area **2604** and combines with the cleaning element area **2610** to create a cleaning solution (and/or sanitizer solution). In this example, the adapter **2602** may contain enough cleaning elements in cleaning element area **2610** for a single use, two uses, three uses, etc. However, in a preferred embodiment, the adapter **2602** is a single use device. In addition, one or more fitting areas **2612** would not be cleaned during a cleaning operation that did not include the adapter **2602**.

[0163] FIG. **26**B shows the fitting device **2611** and the adapter **2602** connected from the cleaning and/or sanitizing operation. In this example, there is no cleaning element area **2610**. Therefore, the cleaning and/or sanitizer would enter the inlet area **2604** to clean and/or sanitize the fitting device **2611** and/or the one or more fitting areas **2612**.

[0164] In FIG. **27**A, an illustration of a dispensing device **2700** is shown. In one example, the dispensing device **2700** may be coupled to an H.sub.2O source **2702** (and/or any other liquid source (e.g., carbonated water, etc.), a tank **2704** (e.g., carbonated tank, nitrogen tank, etc.), a recirculation pump **2706**, and/or one or more lines **2708**. Further, the dispensing device **2700** may include one or more dispensing areas **2710** and/or one or more cooling lines **2712**.

[0165] In FIG. **27**B, another illustration of a dispensing device **2700** is shown, according to various embodiments. In this example, dispensing device **2700** may include one or more input lines **2720**, one or more dispensing areas **2722**, and/or an ice tray **2724**.

[0166] In FIG. **28**, an illustration of an ice rack **2800** is shown, according to one embodiment. In one example, ice rack **2800** may include a metal ice rack **2802** (and/or a plastic and/or glass and/or wood and/or any other material), a filter **2804** (e.g., made of metal, plastic, wood, etc.), a center line **2806**, a left side of a bottom **2808** (and a right side of the bottom), a left center section **2810**, a bucket base **2812** (e.g., metal, plastic, wood, etc.), a wrap base **2814** (e.g., metal, plastic, wood, etc.), a drain out area **2816**, and/or one or more fins **2818**. In one example, the metal ice rack **2802** has three circular patterns where a first circular pattern is under a first dispensing area; a second circular pattern is under an ice dispensing area; and a third circular pattern is under a second dispensing area.

[0167] In another example, the filter **2804** has three circular patterns where a first circular pattern is under a first dispensing area; a second circular pattern is under an ice dispensing area; and a third circular pattern is under a second dispensing area. In addition, the center line **2806** is in the middle of the ice rack **2800** and may be positioned at the center of the ice dispensing area. In another example, the left side of the bottom **2808** and a right side of the bottom may have a plurality of holes for drainage. In addition, the bottom of the ice rack **2800** which includes the left side of the bottom **2808** and a right side of the bottom may be angled downwardly from the center line **2806** in both directions (e.g., left and right). This angle may be 1 degree, 2 degrees, 3, degrees, 4, degrees, 5 degrees, 6 degree, 7 degrees, 8, degrees, 9, degrees, 10 degrees, 11 degrees, 12 degrees, 13, degrees, 14, degrees, 15 degrees, 16 degrees, 17 degrees, 18, degrees, 19, degrees, and/or 20 degrees, according to various embodiments. In another example, the left center section **2810** and/or the right center section may not have any holes (and/or is a solid piece of material). Further, the left center section **2810** and/or the right center section may be positioned over the drain out area **2816**, and/or one or more fins **2818**. Based on the left center section **2810** and/or the right center section not having a holes and/or drainage areas, the drain out area **2816**, and/or one or more fins **2818** are less likely to become clogged because ice, water, waste, and/or any other material cannot fail on top of the drain out area **2816**, and/or one or more fins **2818**. In another example, the one or more fins **2818** stops debris from clogging up the drain out area **2816**.

[0168] In FIG. **29**, an illustration of a cooling system **2900** is shown, according to one embodiment. In one example, the cooling system **2900** may include a carbonated water input line **2902** which goes through various heat exchangers to cool down the carbonated water. In this example, the carbonated water exits the various heat exchangers at a carbonated water exit area **2904**.

[0169] In FIG. **30**, another illustration of a cooling system **3000** is shown, according to one embodiment. In one example, the cooling system **3000** may include a first carbonated water out area **3002** which goes to a recirculation pump. In addition, the cooling system **3000** may include a second carbonated water out area **3004** which goes to one or more valves. In another example, a cooling system **3100** shown in FIG. **31** may include a plain water input area **3102**.

[0170] In FIG. **32**, an illustration of an ice rack **3202** is shown, according to one embodiment. In this example, the ice rack **3202** may include a plurality of holes **3206** (and/or drainage areas) on a right side and a left side of a filter bottom. In addition, a left side and a right side of a center line **3204** may have no holes and/or drainage areas to protect a drainage line **3208**. In one example, the left side and the right side of the center line **3204** protects the drainage line **3208** and/or one more fins **2818** (see FIG. **28**) because the left side and right side of the center line **3204** do not allow ice, water, and debris to built up at the drainage line **3208**.

[0171] In FIG. **33**, another illustration of an ice rack **3300** is shown, according to one embodiment. In one example, ice rack **3300** may include a left side **3302** of a filter, a right side **3304** of the filter, and a center line **3306** located between the left side **3302** and the right side **3304**. In another example, the left side **3302** of the filter and the right side **3304** of the filter may have a plurality of holes (See FIG. **32**) for drainage. In addition, the left side **3302** of the filter and the right side **3304** of the filter may be angled downwardly from the center line **3306** in both directions (e.g., left and right). This angle may be 1 degree, 2 degrees, 3, degrees, 4, degrees, 5 degrees, 6 degree, 7 degrees, 8, degrees, 9, degrees, 10 degrees, 11 degrees, 12 degrees, 13, degrees, 14, degrees, 15 degrees, 16 degrees, 17 degrees, 18, degrees, 19, degrees, and/or 20 degrees, according to various embodiments. In another example, the left center section **2810** and/or the right center section may not have any holes (and/or is a solid piece of material) (See FIG. **28**). Further, the left center section **2810** and/or the right center section may be positioned over the drain out area **2816**, and/or one or more fins **2818**. Based on the left center section **2810** and/or the right center section not having a holes and/or drainage areas, the drain out area **2816**, and/or one or more fins **2818** are less likely to become clogged because ice, water, waste, and/or any other material cannot fail on top of the drain out area **2816**, and/or one or more fins **2818**. In another example, the one or more fins **2818** stop debris from clogging up the drain out area **2816** (See FIG. **28**). In another example, the ice rack **3300** may include one or more drainage lines **3308** and/or one or more fins **3310**.

[0172] In FIG. **34**, an illustration of a user interface **3400** for a dispensing system is shown, according to one embodiment. In one example, the user interface **3400** may include a first display **3402** with a first display area **3406** and a second display **3404** with a second display area **3408**. In one example, the first display **3402** and the second display **3404** are on one display device. In another example, the first display **3402** and the second display **3404** are on two or more display devices. In one example, the first display area **3406** may include a first type of icon **3410**, a second type of icon **3412**, a third type of icon **3414**, a fourth type of icon **3416**, a fifth type of icon **3418**, and an Nth type of icon **3420**. In various embodiments, each different type of icon may indicate a different matter. For example, the first type of icon **3410** may indicate and/or be associated with the drinks with the most sales volume (e.g., the top 5, , , , the top 10, etc.). In another example, the second type of icon **3412** may indicate and/or be associated with a specialty drink (e.g., limited time offer-the purple dream). Further, the third type of icon **3414** may indicate and/or be associated with non-sugar drinks. In addition, the fourth type of icon **3416** may indicate and/or be associated with flavor shots (e.g., cherry, lime, etc.). In another example, the fifth type of icon **3418** may indicate and/or be associated with an energy booster (e.g., caffeine, protein, vitamins, etc.). In another example, the sixth type of icon **3420** may indicate and/or be associated with drinks with average and/or low sales volumes.

[0173] In another example, the second display area **3408** may include a logo icon **3424**, a political drink icon **3426**, a sports team drink icon **3428**, an actor and/or celebrity drink icon **1330**, a regional drink icon **3432**, and/or a national drink icon **3434**. In one example, the logo icon **3424** can be utilized to highlight one or more company brands. In another example, the political drink icon **3426** can be utilized to highlight one or more political parties and/or personal and/or themes and/or messages. For example, if you support proposition X, then drink Y. If you don't support proposition X, then drink Z. In another example, the sports team drink icon **3428** can be utilized to highlight one or more teams, one or more players, one or more mascots, etc. For example, if you think Team Y is going to win the big game, then drink A. If you think Team Z is going to win the big game, then drink B. In another example, the actor and/or celebrity drink icon **1330** may be utilized to highlight one or more actors and/or one or more celebrities. For example, if you like Bob Doe, in once upon a field (or just in general), then drink C. In another example, the regional drink icon **3432** may be used to highlight a local drink preference. For example, City X drinks ZZ almost 80 percent of the time. You should drink ZZ too. In another example, the a national drink icon **3434** may be used to highlight a national drink preference. For example, 65 percent of American drink GG, you should drink GG too.

[0174] In FIGS. **35**A-**35**B, illustrations of a CFiVe Switch cap are shown, according to various embodiments. In one example, the CFiVe switch cap can be used to determine the presence of pressure, and thus product that is at the inlet of the CFiVe flow control valve. This is accomplished by using a microswitch which is actuated with a plunger by the movement of the retainer assembly. In one example, the switch is retained in the cap. Further, the switch cap can be used in addition to or in place of a pressure switch as a sold out feature and/or as a means of detecting actuation downstream of an open valve.

[0175] In FIG. **36**, an illustration of a dispensing system **3600** is shown, according to one embodiment. In one example, the dispensing system **3600** may include a first drink selection area **3602** (e.g., high % Core) and a second drink selection area (e.g., low % recipe). In addition, a drink size area **3606** may be include in the dispensing system **3600**. In another example, one or more functional buttons **3608** may be utilized for a scrolling function, one or more flavor shots, one or more energy shots, one or more additives, and/or any combination thereof. In another example, the dispensing system **3600** may include one or more drink dispensing areas and/or ice dispensing areas **3610**. Further, an ice rack **3612** may be utilized as previously described in this disclosure.

[0176] In FIG. **37**, an illustration of a ventless valve **3700** is shown, according to one embodiment. In one example, ventless valve **3700** may include one or more holes **3702** and one or more closed areas **3704**. In one example, the use of a ventless valve (e.g., a ventless CF Valve) has advantages when a chemical mixing and/or beverage dispensing function is utilized. For example, if there is a diaphragm failure there is no leak from the valve. In this example, any discharge would be downstream through the plumbing into the drain. Further in the case of a solenoid-controlled system, no leak because the solenoid would stay closed and not allow any flow. Further, this ventless valve may produce a sold-out closed valve situation. In one example, as the fluid (liquid and/or gas) leaks through the diaphragm into the closed spring cup/cap the fluid pressure will equalize on either side of the diaphragm and the fluid on the cap side of the diaphragm will act with the spring to hold the diaphragm in the sealing ring in a closed position. This eliminates any flow through the valve. This will eliminate issues with a tear and/or puncture in the diaphragm which is the most common source of failure.

[0177] FIGS. **38**A-**38**B show illustrations of a ventless CF Valve, according to various embodiments. In an example, a valve **3806**A is coupled to a flow path **3800**. In this example, a vent

3802 is utilized in the valve **3806**A. In an embodiment shown in FIG. **38**B, a valve **3806**B is coupled to the flow path **3800**; however, valve **3806**B does not have a vent **3804** which provide all the benefits noted above for a ventless valve.

[0178] In FIG. **39**, an illustration of pressures which can be utilized with the ventless CF Valve is shown; according to various embodiments. In various examples, one or more standard pressures can be achieved with a vented valve. The standard pressures include 7.5, 14, 21, 29, 43, and 58. However, the ventless valve can achieve other pressures, which include 10, 16, 23, 32, 46, and 62. In various other examples, the new pressures may be 10, 14, 18, 20, 23, and/or 27. The pressures are in PSI.

[0179] FIGS. **40**A-**40**B show illustrations of a ventless CF Valve, according to various embodiments. In an example, a valve **4000** is coupled to a flow path. In this example, a vent **4004** is utilized in the valve **4000**. In an embodiment shown in FIG. **40**B, a valve **4002** is coupled to the flow path; however, valve **4002** does not have a vent **4006** which provide all the benefits noted above for a ventless valve.

[0180] FIGS. **41**A-**41**B show illustrations of a ventless CF Valve, according to various embodiments. In an example, a valve **4100** is coupled to a flow path. In this example, a vent **4104** is utilized in the valve **4100**. In an embodiment shown in FIG. **41**B, a valve **4102** is coupled to the flow path; however, valve **4102** does not have a vent **4106** which provide all the benefits noted above for a ventless valve.

[0181] FIGS. **42**A-**42**B show illustrations of a ventless CF Valve, according to various embodiments. In an example, a valve **4200** is coupled to a flow path. In this example, a vent **4204** is utilized in the valve **4200**. In an embodiment shown in FIG. **42**B, a valve **4202** is coupled to the flow path; however, valve **4202** does not have a vent **4206** which provide all the benefits noted above for a ventless valve.

[0182] FIGS. **43**A-**43**B show illustrations of a ventless CF Valve, according to various embodiments. In an example, a valve **4300** is coupled to a flow path. In this example, a vent **4304** is utilized in the valve **4300**. In an embodiment shown in FIG. **43**B, a valve **4302** is coupled to the flow path; however, valve **4302** does not have a vent **4306** which provide all the benefits noted above for a ventless valve.

[0183] FIGS. **44**A-**44**B show illustrations of a ventless CF Valve, according to various embodiments. In an example, a valve **4400** is coupled to a flow path. In this example, a vent **4404** is utilized in the valve **4400**. In an embodiment shown in FIG. **44**B, a valve **4402** is coupled to the flow path; however, valve **4402** does not have a vent **4406** which provide all the benefits noted above for a ventless valve.

[0184] The dispensing system and/or dispensing devices can be utilized with OJ, juices, dairy products, soft drinks, coffee, beer, wine, seltzer, and/or any components thereof (e.g., hops, alcohol, pulp, cream, flavors, syrups, etc.).

[0185] This disclosure relates generally to fluid valves, and is concerned in particular with a regulating valve that is normally closed, that is opened by a variable fluid pressure above a selected threshold level, and that when open, serves to deliver the fluid at a constant pressure and flow rate.

[0186] In one example, a regulating valve for receiving fluid at a variable pressure from a fluid source and for delivering the fluid at a substantially constant pressure and flow rate to a fluid applicator or the like, the valve including: a cup-shaped base having a cylindrical wall segment terminating in an upper rim, and an externally projecting first flange; a cap having an inwardly projecting ledge and an externally projecting second flange, the cup-shaped base and the cap being configured and dimensioned for assembly as a unitary housing, with the cylindrical wall segment of the cup-shaped base inserted into the cap, and with the extent of such insertion being limited by the abutment of the first flange with the second flange to thereby provide a space between the upper rim of the cup-shaped base and the inwardly projecting ledge of the cap; a barrier wall subdividing the interior of the housing into a head section and a base section; a modulating assembly subdividing the base section into a fluid chamber and a spring chamber; an inlet in the cap for connecting the head section to the fluid source; a port in the barrier wall connecting the head section to the fluid chamber, the port being aligned with a central first axis of the valve; an outlet in the cap communicating with the fluid chamber, the outlet being aligned on a second axis transverse to the first axis; a stem projecting from the modulating assembly along the first axis through the port into the head section; a flexible diaphragm supporting the modulating assembly within the housing for movement in opposite directions along the first axis, the diaphragm having an outer periphery captured in the space between the inwardly projecting ledge of the cap and a rim of the cylindrical wall segment of the cup-shaped base; a spring in the spring chamber, the spring being arranged to resiliently urge the modulating assembly into a closed position at which the diaphragm is in sealing contact with the barrier wall to thereby prevent fluid flow from the head section via the port and fluid chamber to the outlet, the spring acting in concert with the modulating assembly and the stem projecting therefrom to modulate the size of the port as an inverse function of the variable fluid pressure in the input sections whereby the pressure and flow rate of the fluid delivered to the outlet is maintained substantially constant, the valve being automatically actuated when the pressure of the fluid acting on the modulating assembly exceeds a threshold level, and being automatically closed when the pressure drops below the threshold level.

[0187] In one embodiment, a cleaning system may include: a CF Valve, the CF Valve coupled to a pressure source and a cleaning unit; the cleaning unit including a cleaning material bag and an outlet area, a cleaning solution recipe generated via the pressure source, the CF Valve, and the cleaning material bag; and a dispensing device coupled to the outlet area which receives a cleaning solution generated by the cleaning solution recipe.

[0188] In addition, the cleaning system may include: a dispensing device outlet area; a solenoid coupled to the CF Valve and the cleaning unit; a solenoid coupled to the CF Valve and the cleaning unit and one or more sensors; and/or one or more sensors. Further, a first sensor is configured to measure temperature data. In addition, the cleaning solution recipe is modified based on the measured temperature data. In addition, the CF Valve may include a housing having axially aligned inlet and outlet ports adapted to be connected respectively to the variable fluid supply and the fluid outlet; a diaphragm chamber interposed between the inlet and the outlet ports, the inlet port being separated from the diaphragm chamber by a barrier wall, the barrier wall having a first passageway extending therethrough from an inner side facing the diaphragm chamber to an outer side facing the inlet port; a cup contained within the diaphragm chamber, the cup having a cylindrical side wall extending from a bottom wall facing the outlet port to a circular rim surrounding an open mouth facing the inner side of the barrier wall, the cylindrical side and bottom walls of the cup being spaced inwardly from adjacent interior surfaces of the housing to define a second passageway connecting the diaphragm chamber to the outlet port; a resilient disc-shaped diaphragm closing the open mouth of the cup, the diaphragm being axially supported by the circular rim and having a peripheral flange overlapping the cylindrical side wall; a piston assembly secured to the center of the diaphragm, the piston assembly having a cap on one side of the diaphragm facing the inner side of the barrier wall, and a base suspended from the opposite side of the diaphragm and projecting into the interior of the cup; a stem projecting from the cap through the first passageway in the barrier wall to terminate in a valve head, the valve head and the outer side of the barrier wall being configured to define a control orifice connecting the inlet port to the diaphragm chamber via the first passageway; and a spring device in the cup coacting with the base of the piston assembly for resiliently urging the diaphragm into a closed position against the inner side of the barrier wall to thereby prevent fluid flow from the inlet port via the first passageway into the diaphragm chamber, the spring device being responsive to fluid pressure above a predetermined level applied to the diaphragm via the inlet port and the first passageway by accommodating movement of the diaphragm away from the inner side of the barrier wall, with the valve head on the stem being moved to adjust the size of the control orifice, thereby maintaining a constant flow of fluid from the inlet port through the first and second passageways to the outlet port for delivery to the fluid outlet.

[0189] Further, the CF Valve may maintain a relative constant flow of fluid from a variable pressure fluid supply to a fluid outlet, the CF Valve including: a) a valve housing having an inlet port and an outlet port adapted to be connected to the variable pressure fluid supply and the fluid outlet; b) a diaphragm chamber interposed between the inlet port and the outlet port; c) a cup contained within the diaphragm chamber; d) a diaphragm closing the cup; e) a piston assembly secured to a center of the diaphragm, the piston assembly having a cap and a base; f) a stem projecting from the cap through a first passageway in a barrier wall to terminate in a valve head; and g) a spring in the cup coacting with the base of the piston assembly for urging the diaphragm into a closed position, and the spring being responsive to fluid pressure above a predetermined level to adjust a size of a control orifice.

[0190] In another example, the CF Valve may maintain a relative constant flow of fluid from a variable pressure fluid supply to a fluid outlet, the CF Valve including: a base having a wall segment terminating in an upper rim, and a projecting first flange; a cap having a projecting ledge and a projecting second flange, the wall segment of the base being located inside the cap with a space between the upper rim of the base and the projecting ledge of the cap; a barrier wall subdividing an interior of a housing into a head section and a base section; a modulating assembly subdividing the base section into a fluid chamber and a spring chamber; an inlet in the cap for connecting the head section to a fluid source; a port in the barrier wall connecting the head section to the fluid chamber, the port being aligned with a central first axis of the CF Valve; an outlet in the cap communicating with the fluid chamber, the outlet being aligned on a second axis transverse to the first axis; a stem projecting from the modulating assembly along the first axis through the port into the head section; a diaphragm supporting the modulating assembly within the housing for movement in opposite directions along the first axis, a spring in the spring chamber, the spring being arranged to urge the modulating assembly into a closed position at which the diaphragm is in sealing contact with the barrier wall, and the spring being responsive to fluid pressure above a predetermined level to adjust a size of a control orifice.

[0191] In another embodiment, a cleaning system may include: a first CF Valve configured to be utilized with a flush function; a second CF Valve configured to be utilized with a clean function; a third CF Valve configured to be utilized with a dwell function; a fourth CF Valve configured to be utilized with a sanitize function; and a controller configured to initiate one or more cleaning recipes.

[0192] In various examples, the controller may initiate a first cleaning recipe including a first flush function, a clean function, a first dwell function, a sanitize function, a second dwell function, and a second flush function. In addition, the controller may initiate a second cleaning recipe including a first flush function, a clean function, a dwell function, a sanitize function, and a second flush function. Further, the controller may initiate a third cleaning recipe including a first flush function, a clean function, a second flush function, a sanitize function, a third flush function, a hold function, and a fourth flush function. In another example, the controller may initiate a fourth cleaning recipe including a first flush function, a clean function, a dwell function, and a second flush function. In addition, the controller may initiate a fifth cleaning recipe including a first flush function, a sanitize function, a dwell function, and a second flush function. Further, the controller may initiate a sixth cleaning recipe including a clean function, a dwell function, and a flush function. In another example, the controller may initiate a seventh cleaning recipe including a flush function, a clean function, and a dwell function.

[0193] A constant flow regulating valve includes a closure mechanism configured and arranged to override the modulating mode of the valve and to close the valve at fluid inlet pressures both below and above the valve's threshold level. The closure mechanism may be selectively deactivated to thereby allow the valve to assume its normal pressure responsive regulating functions. Embodiments of the regulating valve incorporate pressure relief devices and vent seals, with

configurations suitable for incorporation into the trigger assemblies of portable sprayers.
[0194] This disclosure relates generally to fluid valves, and is concerned in particular with a regulating valve that operates in response to a variable fluid inlet pressure above a selected threshold level to deliver the fluid at a constant outlet pressure and flow rate. A closure mechanism is selectively operable either to accommodate the valve's normal pressure responsive regulating functions, or to override such functions by maintaining the valve in a closed state at inlet pressures both above and below the threshold level.
[0195] In one example, valves are normally closed in response to fluid inlet pressures below a threshold level, and operate in a modulating mode in response to variable fluid inlet pressures above the threshold level to deliver fluids at constant outlet pressures and flow rates. However, at fluid inlet pressures above the threshold level, such valves remain open and cannot serve as shut off valves, thus making it necessary to employ additional and separately operable valves to achieve this added function.
[0196] In accordance with one aspect of the present disclosure, the known regulating valves are modified to include closure mechanisms configured and arranged to override the modulating mode of the valves and to maintain closure of the valves at fluid inlet pressures both below and above the threshold level. The closure mechanisms may be selectively deactivated to thereby allow the valves to assume their normal pressure responsive regulating functions.
[0197] In accordance with still another aspect of the present disclosure, the vent opening communicating with the valve's spring chamber is provided with a seal which allows air to escape and enter the spring chamber, but which prevents the escape of liquid from the spring chamber in the event that the valve diaphragm is breached.
[0198] In accordance with another aspect of the present disclosure, a pressure relief mechanism is provided for relieving residual fluid inlet pressure below the threshold level when the valve is closed.
[0199] In one embodiment, a dispensing system may include: a display screen may display one or more icons, the one or more icons including at least a first icon, a second icon, and a third icon, the display screen may transmit one or more signals to one or more processors based on one or more user contacts with the display screen; the one or more processors may receive at least a first sales data, a second sales data, and a third sales data, where the one or more processors may: modify a first size of the first icon based on a first sales data for a first element relating to the first icon; modify a second size of the second icon based on a second sales data for a second element relating to the second icon; and modify a third size of the third icon based on a third sales data for a third element relating to the third icon; and a first dispensing unit which may dispense one or more elements based on the one or more signals.
[0200] In another embodiment, the first size of the first icon is larger than the second size of the second icon based on the first sales data being larger than the second sales data. In another example, the one or more processors may change the first element relating to the first icon to a fourth element and utilize a fourth sales data to determine a fourth size of the first icon. In another example, the one or more processors may modify at least one of first icon, the second icon, and the third icon based on receiving a time-of-day signal. In another example, the one or more processors may modify at least one of first icon, the second icon, and the third icon based on receiving an event signal. In another example, the one or more processors may modify at least one of first icon, the second icon, and the third icon based on receiving a promotional signal. In another example, the dispensing system may include a CF Valve in the first dispensing unit. In another example, the CF Valve includes a housing having axially aligned inlet and outlet ports adapted to be connected respectively to the variable fluid supply and the fluid outlet; a diaphragm chamber interposed between the inlet and the outlet ports, the inlet port being separated from the diaphragm chamber by a barrier wall, the barrier wall having a first passageway extending therethrough from an inner side facing the diaphragm chamber to an outer side facing the inlet port; a cup contained within the

diaphragm chamber, the cup having a cylindrical side wall extending from a bottom wall facing the outlet port to a circular rim surrounding an open mouth facing the inner side of the barrier wall, the cylindrical side and bottom walls of the cup being spaced inwardly from adjacent interior surfaces of the housing to define a second passageway connecting the diaphragm chamber to the outlet port; a resilient disc-shaped diaphragm closing the open mouth of the cup, the diaphragm being axially supported by the circular rim and having a peripheral flange overlapping the cylindrical side wall; a piston assembly secured to the center of the diaphragm, the piston assembly having a cap on one side of the diaphragm facing the inner side of the barrier wall, and a base suspended from the opposite side of the diaphragm and projecting into the interior of the cup; a stem projecting from the cap through the first passageway in the barrier wall to terminate in a valve head, the valve head and the outer side of the barrier wall being configured to define a control orifice connecting the inlet port to the diaphragm chamber via the first passageway; and a spring device in the cup coacting with the base of the piston assembly for resiliently urging the diaphragm into a closed position against the inner side of the barrier wall to thereby prevent fluid flow from the inlet port via the first passageway into the diaphragm chamber, the spring device being responsive to fluid pressure above a predetermined level applied to the diaphragm via the inlet port and the first passageway by accommodating movement of the diaphragm away from the inner side of the barrier wall, with the valve head on the stem being moved to adjust the size of the control orifice, thereby maintaining a constant flow of fluid from the inlet port through the first and second passageways to the outlet port for delivery to the fluid outlet. In another example, the CF Valve is configured to maintain a relative constant flow of fluid from a variable pressure fluid supply to a fluid outlet, the CF Valve including: a) a valve housing having an inlet port and an outlet port adapted to be connected to the variable pressure fluid supply and the fluid outlet; b) a diaphragm chamber interposed between the inlet port and the outlet port; c) a cup contained within the diaphragm chamber; d) a diaphragm closing the cup; e) a piston assembly secured to a center of the diaphragm, the piston assembly having a cap and a base; f) a stem projecting from the cap through a first passageway in a barrier wall to terminate in a valve head; and g) a spring in the cup coacting with the base of the piston assembly for urging the diaphragm into a closed position, and the spring being responsive to fluid pressure above a predetermined level to adjust a size of a control orifice. In another example, the CF Valve is configured to maintain a relative constant flow of fluid from a variable pressure fluid supply to a fluid outlet, the CF Valve including: a base having a wall segment terminating in an upper rim, and a projecting first flange; a cap having a projecting ledge and a projecting second flange, the wall segment of the base being located inside the cap with a space between the upper rim of the base and the projecting ledge of the cap; a barrier wall subdividing an interior of a housing into a head section and a base section; a modulating assembly subdividing the base section into a fluid chamber and a spring chamber; an inlet in the cap for connecting the head section to a fluid source; a port in the barrier wall connecting the head section to the fluid chamber, the port being aligned with a central first axis of the CF Valve; an outlet in the cap communicating with the fluid chamber, the outlet being aligned on a second axis transverse to the first axis; a stem projecting from the modulating assembly along the first axis through the port into the head section; a diaphragm supporting the modulating assembly within the housing for movement in opposite directions along the first axis, a spring in the spring chamber, the spring being arranged to urge the modulating assembly into a closed position at which the diaphragm is in sealing contact with the barrier wall, and the spring being responsive to fluid pressure above a predetermined level to adjust a size of a control orifice. In another example, the first icon and the second icon are on a first side of the display screen and the third icon is on a second side of the display screen. In another example, the dispensing system may include a second dispensing unit, where the one or more processors may transmit a second display signal to the second dispensing unit based on one or more signals from the third icon and to transmit a third display signal to the first dispensing unit based one or more signals from at least one of the first icon and the second icon. In another example, the one or more processors may receive an external signal (via landline, wired, wireless, internet, etc.) to modify at least one of: the first icon, the second icon, the third icon; the first size, the second size, the third size, the first element, the second element, and the third element.

[0201] In one embodiment, a dispensing system may include: a display screen which may display one or more icons, the one or more icons including at least a first icon, a second icon, and a third icon, the display screen may transmit one or more signals to one or more processors based on one or more user contacts with the display screen; the one or more processors may receive at least a first data, a second, and a third data, where the one or more processors may modify: a first position of the first icon based on the first data for a first element relating to the first icon; modify a second position of the second icon based on the second data for a second element relating to the second icon; and modify a third position of the third icon based on the third data for a third element relating to the third icon; and a first dispensing unit may dispense one or more elements based on the one or more signals.

[0202] In another example, the one or more processors may receive an external signal to modify at least one of: the first icon, the second icon, the third icon; the first position, the second position, the third position, the first element, the second element, and the third element. In another example, the first position of the first icon is more prominent than the second position of the second icon based on a comparison of the first data to the second data.

[0203] In another embodiment, a dispensing system may include: a display screen which may display one or more icons, the one or more icons including at least a first icon, a second icon, a third icon, and a fourth icon, the display screen may transmit one or more signals to one or more processors based on one or more user contacts with the display screen; where the first icon has a first size, the second icon has a second size, the third icon has a third size, and the fourth icon has a fourth size; the one or more processors may receive at least a first sales data, a second sales data, a third sales data, and a fourth sales data, where the one or more processors may modify: the first size of the first icon based on the first sales data for a first element relating to the first icon; modify the second size of the second icon based on the second sales data for a second element relating to the second icon; modify the third size of the third icon based on the third sales data for a third element relating to the third icon; and modify the fourth size of the fourth icon based on the fourth sales data for a fourth element relating to the fourth icon; and a first dispensing unit may dispense one or more elements based on the one or more signals; where the modified first size of the first icon is larger than the first size based on the first sales data for the first element increasing versus a first previous sales data.

[0204] In another example, the modified second size of the second icon is smaller than the second size based on the second sales data for the second element decreasing versus a second previous sales data. In another example, the first icon and the second icon are on a first side of the display screen and the third icon and the fourth icon are on a second side of the display screen. In another example, the dispensing system may include a video advertisement and/or a still shot advertisement.

[0205] As used herein, the term “mobile device” refers to a device that may from time to time have a position that changes. Such changes in position may comprise of changes to direction, distance, and/or orientation. In particular examples, a mobile device may comprise of a cellular telephone, wireless communication device, user equipment, laptop computer, other personal communication system (“PCS”) device, personal digital assistant (“PDA”), personal audio device (“PAD”), portable navigational device, or other portable communication device. A mobile device may also comprise of a processor or computing platform adapted to perform functions controlled by machine-readable instructions.

[0206] The methods and/or methodologies described herein may be implemented by various means depending upon applications according to particular examples. For example, such methodologies may be implemented in hardware, firmware, software, or combinations thereof. In a hardware implementation, for example, a processing unit may be implemented within one or more application specific integrated circuits ("ASICs"), digital signal processors ("DSPs"), digital signal processing devices ("DSPDs"), programmable logic devices ("PLDs"), field programmable gate arrays ("FPGAs"), processors, controllers, micro-controllers, microprocessors, electronic devices, other devices units designed to perform the functions described herein, or combinations thereof.

[0207] Some portions of the detailed description included herein are presented in terms of algorithms or symbolic representations of operations on binary digital signals stored within a memory of a specific apparatus or a special purpose computing device or platform. In the context of this particular specification, the term specific apparatus or the like includes a general purpose computer once it is programmed to perform particular operations pursuant to instructions from program software. Algorithmic descriptions or symbolic representations are examples of techniques used by those of ordinary skill in the arts to convey the substance of their work to others skilled in the art. An algorithm is considered to be a self-consistent sequence of operations or similar signal processing leading to a desired result. In this context, operations or processing involve physical manipulation of physical quantities. Typically, although not necessarily, such quantities may take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared or otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to such signals as bits, data, values, elements, symbols, characters, terms, numbers, numerals, or the like. It should be understood, however, that all of these or similar terms are to be associated with appropriate physical quantities and are merely convenient labels. Unless specifically stated otherwise, as apparent from the discussion herein, it is appreciated that throughout this specification discussions utilizing terms such as "processing," "computing," "calculating," "determining" or the like refer to actions or processes of a specific apparatus, such as a special purpose computer or a similar special purpose electronic computing device. In the context of this specification, therefore, a special purpose computer or a similar special purpose electronic computing device is capable of manipulating or transforming signals, typically represented as physical electronic or magnetic quantities within memories, registers, or other information storage devices, transmission devices, or display devices of the special purpose computer or similar special purpose electronic computing device.

[0208] Reference throughout this specification to "one example," "an example," "embodiment," and/or "another example" should be considered to mean that the particular features, structures, or characteristics may be combined in one or more examples. Any combination of any element in this disclosure with any other element in this disclosure is hereby disclosed. For example, an element on pages 2-3 can be combined with any element in this document (e.g., an element from pages 8-9).

[0209] While there has been illustrated and described what are presently considered to be example features, it will be understood by those skilled in the art that various other modifications may be made, and equivalents may be substituted, without departing from the disclosed subject matter. Additionally, many modifications may be made to adapt a particular situation to the teachings of the disclosed subject matter without departing from the central concept described herein. Therefore, it is intended that the disclosed subject matter not be limited to the particular examples disclosed.

Claims

1. A dispensing system comprising: a display screen configured to display one or more icons, the one or more icons including at least a first icon, a second icon, and a third icon, the display screen configured to transmit one or more signals to one or more processors based on one or more user contacts with the display screen; the one or more processors configured to receive at least a first sales data, a second sales data, and a third sales data, where the one or more processors are configured to: modify a first size of the first icon based on the first sales data for a first element relating to the first icon; modify a second size of the second icon based on the second sales data for a second element relating to the second icon; and modify a third size of the third icon based on the third sales data for a third element relating to the third icon; and a first dispensing unit configured to dispense one or more elements based on the one or more signals.

2. The dispensing system of claim 1, wherein the one or more processors are configured to receive an external signal to modify at least one of: the first icon, the second icon, the third icon; the first size, the second size, the third size, the first element, the second element, and the third element.

3. The dispensing system of claim 1, wherein the first size of the first icon is larger than the second size of the second icon based on the first sales data being larger than the second sales data.

4. The dispensing system of claim 1, wherein the one or more processors are configured to change the first element relating to the first icon to a fourth element and utilize a fourth sales data to determine a fourth size of the first icon.

5. The dispensing system of claim 1, wherein the one or more processors are configured to modify at least one of first icon, the second icon, and the third icon based on receiving a time-of-day signal.

6. The dispensing system of claim 1, wherein the one or more processors are configured to modify at least one of first icon, the second icon, and the third icon based on receiving an event signal.

7. The dispensing system of claim 1, wherein the one or more processors are configured to modify at least one of first icon, the second icon, and the third icon based on receiving a promotional signal.

8. The dispensing system of claim 1, further comprising a CF Valve in the first dispensing unit.

9. The dispensing system of claim 7, wherein the CF Valve includes a housing having axially aligned inlet and outlet ports adapted to be connected respectively to the variable fluid supply and the fluid outlet; a diaphragm chamber interposed between the inlet and the outlet ports, the inlet port being separated from the diaphragm chamber by a barrier wall, the barrier wall having a first passageway extending therethrough from an inner side facing the diaphragm chamber to an outer side facing the inlet port; a cup contained within the diaphragm chamber, the cup having a cylindrical side wall extending from a bottom wall facing the outlet port to a circular rim surrounding an open mouth facing the inner side of the barrier wall, the cylindrical side and bottom walls of the cup being spaced inwardly from adjacent interior surfaces of the housing to define a second passageway connecting the diaphragm chamber to the outlet port; a resilient disc-shaped diaphragm closing the open mouth of the cup, the diaphragm being axially supported by the circular rim and having a peripheral flange overlapping the cylindrical side wall; a piston assembly secured to the center of the diaphragm, the piston assembly having a cap on one side of the diaphragm facing the inner side of the barrier wall, and a base suspended from the opposite side of the diaphragm and projecting into the interior of the cup; a stem projecting from the cap through the first passageway in the barrier wall to terminate in a valve head, the valve head and the outer side of the barrier wall being configured to define a control orifice connecting the inlet port to the diaphragm chamber via the first passageway; and a spring device in the cup coacting with the base of the piston assembly for resiliently urging the diaphragm into a closed position against the inner side of the barrier wall to thereby prevent fluid flow from the inlet port via the first passageway into the diaphragm chamber, the spring device being responsive to fluid pressure above a predetermined level applied to the diaphragm via the inlet port and the first passageway by accommodating movement of the diaphragm away from the inner side of the barrier wall, with the valve head on the stem being moved to adjust the size of the control orifice, thereby maintaining a constant flow of fluid from the inlet port through the first and second passageways to the outlet port for delivery to the fluid outlet.

10. The dispensing system of claim 7, wherein the CF Valve is configured to maintain a relative constant flow of fluid from a variable pressure fluid supply to a fluid outlet, the CF Valve including: a) a valve housing having an inlet port and an outlet port adapted to be connected to the variable pressure fluid supply and the fluid outlet; b) a diaphragm chamber interposed between the inlet port and the outlet port; c) a cup contained within the diaphragm chamber; d) a diaphragm closing the cup; e) a piston assembly secured to a center of the diaphragm, the piston assembly having a cap and a base; f) a stem projecting from the cap through a first passageway in a barrier wall to terminate in a valve head; and g) a spring in the cup coacting with the base of the piston assembly for urging the diaphragm into a closed position, and the spring being responsive to fluid pressure above a predetermined level to adjust a size of a control orifice.

11. The dispensing system of claim 7, wherein the CF Valve is configured to maintain a relative constant flow of fluid from a variable pressure fluid supply to a fluid outlet, the CF Valve including: a base having a wall segment terminating in an upper rim, and a projecting first flange; a cap having a projecting ledge and a projecting second flange, the wall segment of the base being located inside the cap with a space between the upper rim of the base and the projecting ledge of the cap; a barrier wall subdividing an interior of a housing into a head section and a base section; a modulating assembly subdividing the base section into a fluid chamber and a spring chamber; an inlet in the cap for connecting the head section to a fluid source; a port in the barrier wall connecting the head section to the fluid chamber, the port being aligned with a central first axis of the CF Valve; an outlet in the cap communicating with the fluid chamber, the outlet being aligned on a second axis transverse to the first axis; a stem projecting from the modulating assembly along the first axis through the port into the head section; a diaphragm supporting the modulating assembly within the housing for movement in opposite directions along the first axis, a spring in the spring chamber, the spring being arranged to urge the modulating assembly into a closed position at which the diaphragm is in sealing contact with the barrier wall, and the spring being responsive to fluid pressure above a predetermined level to adjust a size of a control orifice.

12. The dispensing system of claim 1, wherein the first icon and the second icon are on a first side of the display screen and the third icon is on a second side of the display screen.

13. The dispensing system of claim 11, further comprising a second dispensing unit, where the one or more processors are configured to transmit a second display signal to the second dispensing unit based on one or more signals from the third icon and to transmit a third display signal to the first dispensing unit based one or more signals from at least one of the first icon and the second icon.

14. A dispensing system comprising: a display screen configured to display one or more icons, the one or more icons including at least a first icon, a second icon, and a third icon, the display screen configured to transmit one or more signals to one or more processors based on one or more user contacts with the display screen; the one or more processors configured to receive at least a first data, a second data, and a third data, where the one or more processors are configured to: modify a first position of the first icon based on the first data for a first element relating to the first icon; modify a second position of the second icon based on the second data for a second element relating to the second icon; and modify a third position of the third icon based on the third data for a third element relating to the third icon; and a first dispensing unit configured to dispense one or more elements based on the one or more signals.

15. The dispensing system of claim 1, wherein the one or more processors are configured to receive an external signal to modify at least one of: the first icon, the second icon, the third icon; the first position, the second position, the third position, the first element, the second element, and the third element.

16. The dispensing system of claim 1, wherein the first position of the first icon is more prominent than the second position of the second icon based on a comparison of the first data to the second data.

17. A dispensing system comprising: a display screen configured to display one or more icons, the one or more icons including at least a first icon, a second icon, a third icon, and a fourth icon, the display screen configured to transmit one or more signals to one or more processors based on one or more user contacts with the display screen; where the first icon has a first size, the second icon has a second size, the third icon has a third size, and the fourth icon has a fourth size; the one or more processors configured to receive at least a first sales data, a second sales data, a third sales data, and a fourth sales data, where the one or more processors are configured to: modify the first size of the first icon based on the first sales data for a first element relating to the first icon; modify the second size of the second icon based on the second sales data for a second element relating to the second icon; modify the third size of the third icon based on the third sales data for a third element relating to the third icon; and modify the fourth size of the fourth icon based on the fourth sales data for a fourth element relating to the fourth icon; and a first dispensing unit configured to dispense one or more elements based on the one or more signals; wherein the modified first size of the first icon is larger than the first size based on the first sales data for the first element increasing versus a first previous sales data.

18. The dispensing system of claim 17, wherein the modified second size of the second icon is smaller than the second size based on the second sales data for the second element decreasing versus a second previous sales data.

19. The dispensing system of claim 17, wherein the first icon and the second icon are on a first side of the display screen and the third icon and the fourth icon are on a second side of the display screen.

20. The dispensing system of claim 17, further comprising at least one of a video advertisement and a still shot advertisement.